Volume I

Logic, Sets, and Proof

Set Theory

Self-Study Notes

William Sollers

June 26, 2026

From Cantor to Ito

Front Matter

Series. From Cantor to Ito

Volume. Volume I: Logic, Sets, and Proof

Book. Set Theory

Contents

Chapter 1

Set Theory

set-theory

Predicate Logic → **Set Theory** → Relations

1.1 Model of Set Theory

Theory (ZF, ZFC)

Depends on: `theory:first-order-logic`

Language

$$\mathcal{L}_{\in} = \{\in\}$$

First-order language with equality. The symbol $\in$ is binary. Equality is logical.

Presented Theory

$$\mathrm{ZF} := \mathrm{Cn}_{\vdash}(\Sigma_{\mathrm{ZF}}), \qquad \mathrm{ZFC} := \mathrm{Cn}_{\vdash}\big(\Sigma_{\mathrm{ZF}} \cup \{\mathrm{AC}\}\big).$$

Axioms and Laws

1. **1. Extensionality**

$$\forall x\,\forall y\,\big[\forall z\,(z \in x \leftrightarrow z \in y) \rightarrow x = y\big].$$

2. **2. Empty Set**

$$\exists x\,\forall y\,\neg(y \in x).$$

3. **3. Pairing**

$$\forall x\,\forall y\,\exists z\,\forall w\,\big(w \in z \leftrightarrow (w = x \vee w = y)\big).$$

4. **4. Union**

$$\forall x\,\exists y\,\forall z\,\big[z \in y \leftrightarrow \exists w\,(w \in x \wedge z \in w)\big].$$

5. **5. Power Set**

$$\forall x\,\exists y\,\forall z\,\big[z \in y \leftrightarrow \forall w\,(w \in z \rightarrow w \in x)\big].$$

6. **6. Infinity**

$$\begin{aligned}\exists x\,\Big[&\exists y\big(y \in x \wedge \forall w\,\neg(w \in y)\big)\\ &\wedge \forall y\big(y \in x \rightarrow \exists z\,(z \in x \wedge \forall w\,(w \in z \leftrightarrow (w \in y \vee w = y)))\big)\Big].\end{aligned}$$

7. **7. Separation Schema** For each formula $\varphi(z, \bar{w})$ in which y is not free:

$$\forall \bar{w}\,\forall x\,\exists y\,\forall z\,\big[z \in y \leftrightarrow (z \in x \wedge \varphi(z, \bar{w}))\big].$$

8. **8. Replacement Schema** For each formula $\varphi(x, y, \bar{w})$ in which B is not free:

$$\begin{aligned}\forall \bar{w}\,\forall A\,\Big[&\forall x\big(x \in A \rightarrow \exists! y\,\varphi(x, y, \bar{w})\big)\\ &\rightarrow \exists B\,\forall y\big(y \in B \leftrightarrow \exists x\,(x \in A \wedge \varphi(x, y, \bar{w}))\big)\Big].\end{aligned}$$

9. **9. Foundation**

$$\forall x\,\Big[\exists y\,(y \in x) \rightarrow \exists y\big(y \in x \wedge \neg\exists z\,(z \in x \wedge z \in y)\big)\Big].$$

Layer Delta

AC

$$\begin{aligned}\forall x\,\Big[&\big(\forall y\,(y \in x \rightarrow \exists z\; z \in y)\\ &\wedge \forall y\,\forall y'\,((y \in x \wedge y' \in x \wedge y \neq y') \rightarrow \neg\exists z\,(z \in y \wedge z \in y'))\big)\\ &\rightarrow \exists c\,\forall y\,(y \in x \rightarrow \exists! z\,(z \in y \wedge z \in c))\Big].\end{aligned}$$

Model (Set Theory)

Model 1.1.1 (Set Theory Model)**. Depends on:** `model:first-order-logic; bridge:free-sigma-algebra-evaluation; theory:set-theory; bridge:consequence`

Structure

$$\mathcal{A} = (A, E), \qquad A \neq \varnothing, \qquad E \subseteq A \times A.$$
$$a \; E \; b \quad :\Longleftrightarrow \quad (a, b) \in E.$$
$$\mathcal{A} = \langle A, I \rangle, \qquad I(\in) = E.$$

Sorts and Domains

A universe

Interpretation

Constants. None.

Interpretation

Functions. None.

Relations

E $E \subseteq A \times A$
Interpretation of $\in$.

Layer Delta

No constant or function symbols occur in $\mathcal{L}_{\in}$, so every term is a variable and FOL term evaluation reduces to $[\![x]\!]_{\mathcal{A},s} = s(x)$.

Syntax–Semantics Bridge

$$\mathcal{A}, s \models x \in y \quad \Longleftrightarrow \quad (s(x), s(y)) \in E.$$

Models of the Theory

$$\mathcal{A} \models \mathrm{ZF} \quad :\Longleftrightarrow \quad \forall \varphi \, (\varphi \in \mathrm{ZF} \to \mathcal{A} \models \varphi).$$
$$\mathcal{A} \models \mathrm{ZFC} \quad :\Longleftrightarrow \quad \mathcal{A} \models \mathrm{ZF} \wedge \mathcal{A} \models \mathrm{AC}.$$

Theorems and Consequences

1. One nonlogical symbol is interpreted: $I(\in) = E$.
2. $\mathrm{Th}(\mathcal{A})$ is inherited from *Consequence*.

Definitional Extension (Set Theory)

Depends on: `defext:first-order-logic; theory:set-theory; model:set-theory`

Layer Delta

$$\begin{aligned}\text{existence} &\leftarrow \text{Empty Set, Pairing, Union, Power Set,}\\ &\qquad\text{Infinity, Separation, Replacement;}\\ \text{uniqueness} &\leftarrow \text{Extensionality;}\\ \text{default value} &\leftarrow \varnothing.\end{aligned}$$

Representative-independence is delegated to quotient-specific extensions.

Instances

$\subseteq$ relation

$$\forall x\,\forall y\,\big(x \subseteq y \leftrightarrow \forall w\,(w \in x \to w \in y)\big).$$

No unique-existence obligation.
$\varnothing$ constant

$$\psi_\varnothing(y) :\equiv \forall w\,\neg(w \in y), \qquad \exists!\ \text{receipt: Empty Set + Extensionality.}$$

$$\forall y\,\big(\varnothing = y \leftrightarrow \forall w\,\neg(w \in y)\big).$$

$\{\cdot,\cdot\}$ binary function

$$\psi(x_1, x_2, y) :\equiv \forall w\,(w \in y \leftrightarrow (w = x_1 \vee w = x_2)).$$

$$\exists!\ \text{receipt: Pairing + Extensionality.}$$

$$\forall x_1\,\forall x_2\,\forall y\,\big(\{x_1, x_2\} = y \leftrightarrow \psi(x_1, x_2, y)\big).$$

$\bigcup$ unary function

$$\psi(x, y) :\equiv \forall z\,(z \in y \leftrightarrow \exists w\,(w \in x \wedge z \in w)).$$

$$\exists!\ \text{receipt: Union + Extensionality.}$$

$$\forall x\,\forall y\,\big(\bigcup x = y \leftrightarrow \psi(x, y)\big).$$

$\mathcal{P}$ unary function

$$\psi(x, y) :\equiv \forall z\,(z \in y \leftrightarrow \forall w\,(w \in z \to w \in x)).$$

$$\exists!\ \text{receipt: Power Set + Extensionality.}$$

$$\forall x\,\forall y\,\big(\mathcal{P}(x) = y \leftrightarrow \psi(x, y)\big).$$

1.2 Sets

1.3 Sets, Membership, and ZFC Axioms

Axiom (Extensionality)

Axiom 1.3.1 (Axiom of Extensionality)**.** Two sets are equal iff they have the same elements.

$$\forall A \forall B \left(A = B \iff \forall x\, (x \in A \leftrightarrow x \in B) \right).$$

Axiom (Empty Set)

Axiom 1.3.2 (Axiom of Empty Set)**.** There exists a set with no elements.

$$\exists A\, \forall x\, (x \notin A).$$

Axiom (Pairing)

Axiom 1.3.3 (Axiom of Pairing)**.** For any two sets, there exists a set containing exactly those two sets.

$$\forall A\, \forall B\, \exists C\, \forall x\, (x \in C \leftrightarrow (x = A \lor x = B)).$$

Axiom (Union)

Axiom 1.3.4 (Axiom of Union)**.** For any family of sets, there exists a set containing exactly the elements of those sets.

$$\forall A\, \exists U\, \forall x\, \left(x \in U \leftrightarrow \exists B\, (B \in A \land x \in B)\right).$$

Axiom (Power Set)

Axiom 1.3.5 (Axiom of Power Set)**.** For any set, there exists a set of all its subsets.

$$\forall A\, \exists P\, \forall x\, (x \in P \leftrightarrow x \subseteq A).$$

Axiom (Infinity)

Axiom 1.3.6 (Axiom of Infinity)**.** There exists an infinite set.

$$\exists A \left(\varnothing \in A \land \forall x\, (x \in A \to x \cup \{x\} \in A) \right).$$

Axiom Schema (Separation)

Axiom 1.3.7 (Axiom Schema of Separation)**.** Given a set and a property, there exists a subset containing exactly the elements satisfying that property. For any formula $\varphi(x)$,

$$\forall A\, \exists B\, \forall x\, (x \in B \leftrightarrow (x \in A \land \varphi(x))).$$

Axiom Schema (Replacement)

Axiom 1.3.8 (Axiom Schema of Replacement)**.** If $\varphi(x, y)$ defines a functional relation on a set A, then the image of A under φ is a set. For any formula $\varphi(x, y)$,

$$\forall A\left(\left(\forall x \in A\, \exists! y\, \varphi(x, y)\right) \to \exists B\, \forall y\, \left(y \in B \leftrightarrow \exists x \in A\, \varphi(x, y)\right)\right).$$

Axiom (Foundation)

Axiom 1.3.9 (Axiom of Foundation)**.** Every nonempty set has an $\in$-minimal element.

$$\forall A\left(A \neq \varnothing \to \exists x \in A\,(x \cap A = \varnothing)\right).$$

Axiom (Choice)

Axiom 1.3.10 (Axiom of Choice)**.** For any family of nonempty sets, there exists a selection function.

$$\forall A\left((\forall B \in A\,(B \neq \varnothing)) \to \exists f\, \forall B \in A\,(f(B) \in B)\right).$$

Definition (Set and Membership)

Definition 1.3.11 (Set and Membership)**.** In axiomatic set theory, the notions of *set* and *membership* are *primitive.*

- A *set* is an object.
- *Membership* is a binary relation, denoted by $\in$, between objects.

If x is an object and A is a set, the statement $x \in A$ is read as "x is an element of A." No definition of "set" or "$\in$" is given in more basic terms. Their meaning is determined entirely by the axioms governing them.

Remark (English reading)*.* Primitive means we do not define these in terms of simpler concepts. Rather, we fix their meaning implicitly by stating the axioms they must obey. This is the standard approach in formal mathematics: rules of behaviour replace informal definitions.

Remark (Consequence)*.* All subsequent notions—subsets, ordered pairs, relations, functions, number systems—are *definitions* introduced within this axiomatic framework. Every proved result is a *theorem* derived logically from the axioms via the rules of inference.

Remark (Role of the axioms)*.* The axioms are not statements to be proved, but rules specifying how $\in$ behaves and which sets are permitted to exist. From this point onward, set theory functions as the underlying language of mathematics: all reasoning about mathematical objects is ultimately grounded in these axioms, even when they are not cited explicitly.

Remark (Separation vs. Replacement)*.* Separation is a schema: one axiom instance per definable property φ. It carves out subsets of an already-existing set. Replacement is strictly stronger: it can produce sets whose elements are not already contained in any pre-existing set, allowing the construction of large stages of the cumulative hierarchy.

Remark (Axiom of Choice and right inverses)**.** The Axiom of Choice is equivalent to many statements used later in analysis, including Zorn's Lemma, the well-ordering theorem, and the statement that every surjective function has a right inverse. It is independent of the other ZFC axioms.

Foundations — Quick Reference

Concept	**Meaning**	**Detail**
Set, membership	Primitive notions; meaning fixed by axioms	↓ Def
Extensionality	Sets equal iff same elements	↓ Ax
Empty Set	Exists a set with no elements	↓ Ax
Pairing	Any two sets can be collected	↓ Ax
Union	Elements of sets in a family	↓ Ax
Power Set	All subsets form a set	↓ Ax
Infinity	An infinite set exists	↓ Ax
Separation	Subsets by definable property	↓ Ax
Replacement	Functional image of a set is a set	↓ Ax
Foundation	No infinite descending $\in$-chains	↓ Ax
Choice	Selection function exists	↓ Ax

Set Operations — Quick Reference

Concept	**Notation**	**Membership condition**	**Detail**
Empty set	$\varnothing$	$x \in \varnothing$ never	↓ Def
Subset	$A \subseteq B$	$x \in A \Rightarrow x \in B$	↓ Def
Proper subset	$A \subsetneq B$	$A \subseteq B$ and $A \neq B$	↓ Def
Set equality	$A = B$	same elements both ways	↓ Def
Union	$A \cup B$	in A or in B	↓ Def
Intersection	$A \cap B$	in A and in B	↓ Def
Set difference	$A \setminus B$	in A but not B	↓ Def
Symmetric diff.	$A \triangle B$	in exactly one of A, B	↓ Def
Complement	$A^c = U \setminus A$	in U but not A	↓ Def
Cartesian product	$A \times B$	all ordered pairs (a, b)	↓ Def
Power set	$\mathcal{P}(A)$	all subsets of A	↓ Def
De Morgan	$(A \cup B)^c = A^c \cap B^c$	complement of union	↓ Thm

1.4 Set Constructions and Operations

Definition (Empty Set)

Definition 1.4.1 (Empty Set)**.** The unique set with no elements is called the *empty set* and is denoted $\varnothing$.

Remark (Dependencies).

- Axiom of Empty Set

Remark (Vacuous truth). Statements of the form $\forall x \in \varnothing,\ P(x)$ are *vacuously true*: there is no element $x \in \varnothing$ for which $P(x)$ could fail. This is not a special convention but a consequence of how universal quantification is defined.

Definition (Subset)

Definition 1.4.2 (Subset)**.** Let A and B be sets. We say A is a *subset* of B, written $A \subseteq B$, if every element of A is also an element of B:

$$A \subseteq B \iff \forall x\,(x \in A \to x \in B).$$

Remark (Dependencies).

- Axiom Schema of Separation
- Universal Formula

Definition (Proper Subset)

Definition 1.4.3 (Proper Subset)**.** A is a *proper subset* of B, written $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Remark (Dependencies).

- Subset

Remark (Notation convention). Some authors write $A \subset B$ for a proper subset; others use it to mean $A \subseteq B$. In these notes, $\subseteq$ always denotes subset (possibly equal) and $\subsetneq$ always denotes proper subset.

Definition (Set Equality)

Definition 1.4.4 (Set Equality)**.** Two sets A and B are *equal*, written $A = B$, if they have the same elements:

$$A = B \iff \forall x\,(x \in A \leftrightarrow x \in B).$$

Equivalently, $A = B \iff (A \subseteq B \wedge B \subseteq A)$.

Remark (Dependencies).

- Subset
- Axiom of Extensionality

Remark (Proof strategy). In practice, to prove $A = B$ one shows mutual inclusion: first $A \subseteq B$ (take arbitrary $x \in A$ and deduce $x \in B$), then $B \subseteq A$. This two-step structure appears throughout set-theoretic proofs.

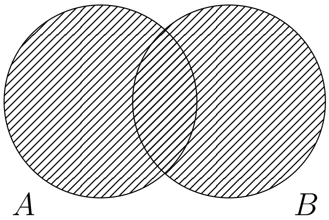
Definition (Union)

Definition 1.4.5 (Union)**.** The *union* of A and B, denoted $A \cup B$, is the set of all elements belonging to at least one of the two sets:

$$A \cup B = \{\, x \mid x \in A \vee x \in B \,\}.$$

Remark (Dependencies).

- Set and Membership
- Axiom Schema of Separation



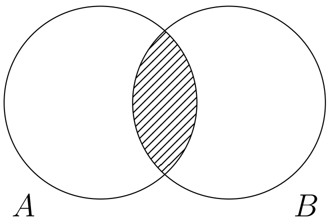
Definition (Intersection)

Definition 1.4.6 (Intersection)**.** The *intersection* of A and B, denoted $A \cap B$, is the set of all elements common to both:

$$A \cap B \;=\; \{\, x \mid x \in A \land x \in B \,\}.$$

Remark (Dependencies).

- Set and Membership
- Axiom Schema of Separation



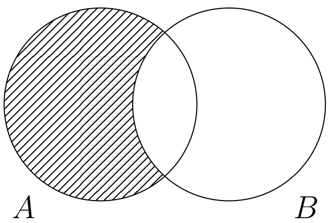
Definition (Set Difference)

Definition 1.4.7 (Set Difference)**.** The *set difference* $A \setminus B$ is the set of elements in A but not in B:

$$A \setminus B \;=\; \{\, x \mid x \in A \land x \notin B \,\}.$$

Remark (Dependencies).

- Set and Membership
- Axiom Schema of Separation



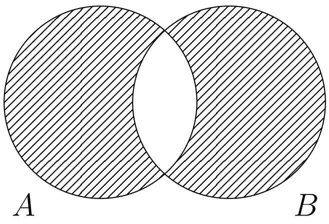
Definition (Symmetric Difference)

Definition 1.4.8 (Symmetric Difference)**.** The *symmetric difference* $A \triangle B$ is the set of elements belonging to exactly one of A and B:

$$A \triangle B \;=\; (A \setminus B) \cup (B \setminus A) \;=\; \{\, x \mid (x \in A \vee x \in B) \wedge x \notin A \cap B \,\}.$$

Remark (Dependencies).

- Set Difference
- Union
- Intersection



Remark (Algebraic structure of $\triangle$)*.* The symmetric difference is commutative and associative. Under $\triangle$, the power set $\mathcal{P}(U)$ forms an abelian group with identity $\varnothing$ and every element its own inverse ($A \triangle A = \varnothing$).

Definition (Complement)

Definition 1.4.9 (Complement)**.** Let U be a fixed universe and $A \subseteq U$. The *complement* of A relative to U, denoted A^c, is:

$$A^c \;=\; U \setminus A.$$

Remark (Dependencies).

- Set and Membership
- Axiom Schema of Separation

Definition (Cartesian Product)

Definition 1.4.10 (Cartesian Product)**.** The *Cartesian product* of sets A and B is:

$$A \times B = \{\, (a, b) \mid a \in A \text{ and } b \in B \,\}.$$

Remark (Dependencies).

- Ordered Pair
- Axiom of Pairing
- Axiom of Power Set
- Axiom Schema of Separation

Remark (Ordered pairs). The ordered pair (a, b) is formally defined as $\{\{a\}, \{a, b\}\}$. This Kuratowski encoding ensures the key property: $(a, b) = (c, d)$ iff $a = c$ and $b = d$.

Remark (Non-commutativity). In general, $A \times B \neq B \times A$. The product is however associative up to canonical isomorphism: $(A \times B) \times C \cong A \times (B \times C)$.

Definition (Power Set)

Definition 1.4.11 (Power Set)**.** The *power set* of A, denoted $\mathcal{P}(A)$, is the set of all subsets of A:

$$\mathcal{P}(A) \ := \ \{ S \mid S \subseteq A \}.$$

Remark (Dependencies).

- Axiom of Power Set

Remark (Size). If A has n elements, then $|\mathcal{P}(A)| = 2^n$. For infinite A, Cantor's theorem shows $|\mathcal{P}(A)| > |A|$ strictly.

Theorem (De Morgan's Laws)

Theorem 1.4.12 (De Morgan's Laws)**.** *Let U be a universe and $A, B \subseteq U$. Then*

$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c.$$

Go to proof.

Remark (Standard quantified statement). $\forall A, B \subseteq U : \ \forall x \left(x \notin A \cup B \leftrightarrow (x \notin A \wedge x \notin B)\right)$, and dually $\forall x \left(x \notin A \cap B \leftrightarrow (x \notin A \vee x \notin B)\right)$.

Remark (Dependencies).

- Union
- Intersection
- Complement

Remark (Connection to propositional logic). De Morgan's laws mirror the propositional logic identities $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$ and $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ under the correspondence $\cap \leftrightarrow \wedge$, $\cup \leftrightarrow \vee$, $A^c \leftrightarrow \neg A$.

Definition (Set Duality)

Definition 1.4.13 (Set Duality)**.** Two set-theoretic expressions over a fixed universe U are *dual* if one is obtained from the other by simultaneously replacing $\cup \leftrightarrow \cap$ and $\varnothing \leftrightarrow U$, with complements unchanged.

Remark (Standard quantified statement). $\forall$ expressions E, E' over U: E and E' are dual iff E' is obtained from E by the substitution $\cup \mapsto \cap$, $\cap \mapsto \cup$, $\varnothing \mapsto U$, $U \mapsto \varnothing$.

Remark (Dependencies).

- Union

- Intersection
- Empty Set
- Complement

Corollary

Corollary 1.4.14 (Principle of Set Duality)**.** *Any identity involving* $\cup$, $\cap$, $\varnothing$, *and* U *that holds for all subsets of a universe remains valid when each operation and constant is replaced by its dual.*

Remark (Standard quantified statement)*.* For any identity $\forall A_1, \ldots, A_n \subseteq U : \ E(A_i, \cup, \cap, \varnothing, U) = E'(A_i, \cup, \cap, \varnothing, U)$, the dual identity $\forall A_1, \ldots, A_n \subseteq U : \ E^d = E'^d$ holds, where E^d replaces $\cup \leftrightarrow \cap$ and $\varnothing \leftrightarrow U$.

Remark (Dependencies)*.*

- Set Duality
- Axiom of Extensionality

Remark (Using duality in proofs)*.* To prove a statement involving unions and intersections, it often suffices to prove one version; the dual statement follows immediately.

Example (Distributive law via duality)*.* The two distributive laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad \text{and} \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

are duals of each other. Proving either one and applying the Principle of Set Duality yields the other immediately.

Remark (Transition to relations)*.* The Cartesian product provides a way to encode ordered information. Relations and functions will be defined as special subsets of Cartesian products, so they require no new foundational objects.

Set Algebra Laws — Quick Reference

Law	**Union form**	**Intersection form**	**Detail**
Commutativity	$A \cup B = B \cup A$	$A \cap B = B \cap A$	↓ Thm
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$	analogous	↓ Thm
Distributivity	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	dual	↓ Thm
Identity	$A \cup \varnothing = A$	$A \cap U = A$	↓ Thm
Absorption	$A \cup (A \cap B) = A$	$A \cap (A \cup B) = A$	↓ Thm
Involution	$(A^c)^c = A$	—	↓ Thm

1.5 Algebraic Laws of Set Operations

Remark (Algebraic structure of set operations)*.* The operations $\cup$, $\cap$, and complement satisfy algebraic laws analogous to those of logical connectives. Together they endow $\mathcal{P}(U)$ with the structure of a *Boolean algebra*. These laws justify manipulation of set expressions in later proofs, particularly those involving equivalence classes, partitions, and functions.

Theorem (Commutativity of Union and Intersection)

Theorem 1.5.1 (Commutativity of Union and Intersection)**.** *Let A, B be sets. Then*

$$A \cup B = B \cup A \quad \textit{and} \quad A \cap B = B \cap A.$$

Go to proof.

Remark (Standard quantified statement). $\forall A, B : \ \forall x\,(x \in A \cup B \leftrightarrow x \in B \cup A)$ and $\forall x\,(x \in A \cap B \leftrightarrow x \in B \cap A)$.

Remark (Dependencies).

- Axiom of Extensionality
- Union
- Intersection

Theorem (Associativity of Union and Intersection)

Theorem 1.5.2 (Associativity of Union and Intersection)**.** *Let A, B, C be sets. Then*

$$(A \cup B) \cup C = A \cup (B \cup C) \quad \textit{and} \quad (A \cap B) \cap C = A \cap (B \cap C).$$

Go to proof.

Remark (Standard quantified statement). $\forall A, B, C : \ \forall x\,\big((x \in (A \cup B) \cup C) \leftrightarrow (x \in A \cup (B \cup C))\big)$, and analogously for $\cap$.

Remark (Dependencies).

- Axiom of Extensionality
- Union
- Intersection

Theorem (Distributive Laws)

Theorem 1.5.3 (Distributive Laws)**.** *Let A, B, C be sets. Then*

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Go to proof.

Remark (Standard quantified statement). $\forall A, B, C : \ \forall x\,\big(x \in A \cap (B \cup C) \leftrightarrow x \in (A \cap B) \cup (A \cap C)\big)$, and dually for $\cup$ over $\cap$.

Remark (Dependencies).

- Axiom of Extensionality
- Union

- Intersection

Theorem (Identity and Absorption Laws)

Theorem 1.5.4 (Identity and Absorption Laws)**.** *Let A, B be sets and U a universe with $A \subseteq U$. Then*

$$A \cup \varnothing = A, \qquad A \cap U = A,$$
$$A \cup (A \cap B) = A, \qquad A \cap (A \cup B) = A.$$

Go to proof.

Remark (Standard quantified statement). $\forall A \subseteq U : \forall x\,(x \in A \cup \varnothing \leftrightarrow x \in A)$ and $\forall x\,(x \in A \cap U \leftrightarrow x \in A)$. Absorption: $\forall A, B : \forall x\,(x \in A \cup (A \cap B) \leftrightarrow x \in A)$, and dually.

Remark (Dependencies).

- Axiom of Extensionality
- Empty Set
- Union
- Intersection
- Subset

Theorem (Involution of Complement)

Theorem 1.5.5 (Involution of Complement)**.** *For any $A \subseteq U$,*

$$(A^c)^c = A.$$

Go to proof.

Remark (Standard quantified statement). $\forall A \subseteq U : \forall x\,(x \in (A^c)^c \leftrightarrow x \in A)$.

Remark (Dependencies).

- Axiom of Extensionality
- Complement

Remark (Non-commutative and non-associative operations). Set difference $\setminus$ is neither commutative nor associative:

$$A \setminus B \neq B \setminus A, \qquad (A \setminus B) \setminus C \neq A \setminus (B \setminus C) \quad \text{in general.}$$

It interacts with union and intersection via:

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C), \qquad A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$$

These follow from $A \setminus B = A \cap B^c$ together with De Morgan's laws.

Remark (Cartesian product). The Cartesian product $\times$ is not commutative, is associative only up to canonical isomorphism, and distributes over union in each coordinate:

$$A \times (B \cup C) = (A \times B) \cup (A \times C).$$

1.6 Families

Indexed Families — Quick Reference

Concept	Meaning	Detail
Indexed family	Function $F : I \to \mathcal{P}(U)$; written $\{A_i\}_{i\in I}$	↓ Def
Indexed union	$\bigcup_{i\in I} A_i$: in at least one A_i	↓ Def
Indexed intersection	$\bigcap_{i\in I} A_i$: in every A_i	↓ Def
Pairwise disjoint	$i \neq j \Rightarrow A_i \cap A_j = \varnothing$	↓ Def
Cover	$\bigcup_{C\in\mathcal{C}} C = A$	↓ Def
Indexed De Morgan	Complements exchange indexed unions and intersections	↓ Thm
Arbitrary product	$\prod_{i\in I} A_i$: choice functions $f : I \to \bigcup A_i$	↓ Def

1.7 Indexed Families of Sets

Remark (Motivation). Many set-theoretic constructions require not just pairs of sets but infinite families. Indexed families provide the formal language for partitions, equivalence classes, unions over countably or uncountably many sets, and products indexed by arbitrary index sets.

Definition (Indexed Family of Sets)

Definition 1.7.1 (Indexed Family of Sets)**.** Let I be a set (the *index set*) and U a universe. An *indexed family of sets* is a function

$$F : I \to \mathcal{P}(U),$$

with $F(i)$ typically written A_i. The family is denoted $\{A_i\}_{i\in I}$.

Remark (Dependencies).

- Function
- Power Set

Remark (Family vs. set of sets). An indexed family is formally a *function*, not a set of sets. Different indices may correspond to the same set: $i \neq j$ does not imply $A_i \neq A_j$. This distinction matters for equivalence class constructions.

Definition (Indexed Union)

Definition 1.7.2 (Indexed Union)**.** The *union* of the indexed family $\{A_i\}_{i\in I}$ is:

$$\bigcup_{i\in I} A_i \;:=\; \{\, x \mid \exists i \in I \text{ such that } x \in A_i \,\}.$$

Remark (Dependencies).

- Indexed Family of Sets
- Union

- Existential Formula

Definition (Indexed Intersection)

Definition 1.7.3 (Indexed Intersection)**.** For $I \neq \varnothing$, the *intersection* of $\{A_i\}_{i\in I}$ is:

$$\bigcap_{i\in I} A_i \;:=\; \{\, x \mid \forall i \in I,\ x \in A_i \,\}.$$

Remark (Dependencies).

- Indexed Family of Sets
- Intersection
- Universal Formula

Remark (Why $I \neq \varnothing$ is required). If $I = \varnothing$, then $\forall i \in I, x \in A_i$ holds vacuously for every x, making the intersection the entire universe U. This is normally left undefined or requires specifying a background universe explicitly.

Theorem (De Morgan's Laws for Indexed Families)

Theorem 1.7.4 (De Morgan's Laws for Indexed Families)**.** *Let $\{A_i\}_{i\in I}$ be an indexed family of subsets of a universe U. Then*

$$U \setminus \bigcup_{i\in I} A_i = \bigcap_{i\in I}(U \setminus A_i).$$

If $I \neq \varnothing$, then

$$U \setminus \bigcap_{i\in I} A_i = \bigcup_{i\in I}(U \setminus A_i).$$

Go to proof.

Remark (Dependencies).

- Indexed Family of Sets
- Indexed Union
- Indexed Intersection
- Complement

Definition (Pairwise Disjoint Family)

Definition 1.7.5 (Pairwise Disjoint Family)**.** The family $\{A_i\}_{i\in I}$ is *pairwise disjoint* if

$$\forall i, j \in I,\ i \neq j \to A_i \cap A_j = \varnothing.$$

Remark (Dependencies).

- Indexed Family of Sets

- Intersection
- Empty Set
- Set Equality

Definition (Cover)

Definition 1.7.6 (Cover)**.** A collection $\mathcal{C} \subseteq \mathcal{P}(A)$ is a *cover* of A if

$$\bigcup_{C \in \mathcal{C}} C = A.$$

Remark (Standard quantified statement). $\mathcal{C}$ covers $A \leftrightarrow \forall x \in A,\ \exists C \in \mathcal{C},\ x \in C$.

Arbitrary Cartesian Products

Definition (Arbitrary Cartesian Product)

Definition 1.7.7 (Arbitrary Cartesian Product)**.** Let $\{A_i\}_{i \in I}$ be an indexed family of sets. The *Cartesian product* is:

$$\prod_{i \in I} A_i := \left\{ f : I \to \bigcup_{i \in I} A_i \;\middle|\; \forall i \in I,\ f(i) \in A_i \right\}.$$

Remark (Dependencies).

- Indexed Family of Sets
- Function
- Indexed Union

Remark (Elements as choice functions). An element of $\prod_{i \in I} A_i$ is a *choice function*: a function that assigns to each index i an element of the corresponding set A_i. For finite $I = \{1, \ldots, n\}$, this reduces to the familiar n-tuple $(a_1, \ldots, a_n)$.

Remark (Axiom of Choice connection). For infinite I, the product $\prod_{i \in I} A_i$ is nonempty iff a choice function exists. This existence is not guaranteed by the other ZFC axioms: it is equivalent to the Axiom of Choice. Thus the Axiom of Choice is precisely the assertion that arbitrary products of nonempty sets are nonempty.

Covers and FIP — Quick Reference

Concept	Meaning	Detail
Cover of A	$\bigcup_{C\in\mathcal{C}} C \supseteq A$	↓ Def
Subcover	Sub-collection of a cover that is still a cover	↓ Def
Finite cover	Cover with finitely many members	↓ Def
Finite subcover	Finite subcollection that is still a cover	↓ Def
Open cover	Cover whose members are all open sets	↓ Def
Finite Intersection Property	Every finite subcollection has nonempty intersection	↓ Def
FIP–cover duality	$\mathcal{C}$ covers A $\iff$ closed complements fail FIP	↓ Prop

1.8 Covers, Subcovers, and the Finite Intersection Property

Compactness is one of the central concepts of analysis, and its definition is stated in terms of covers: a space is compact if every open cover has a finite subcover. This is a purely set-theoretic condition — it makes no reference to distance or continuity. The work of understanding compactness therefore begins here, with the vocabulary of covers. Once these definitions are in place, the definition of compactness will be a single sentence, and the difficulty will lie entirely in the geometric and analytic content, not the set-theoretic framework.

Definition (Cover)

Definition 1.8.1 (Cover)**.** Let A be a set and $\mathcal{C}$ a collection of sets. $\mathcal{C}$ is a *cover* of A if every element of A belongs to at least one member of $\mathcal{C}$:

$$A \subseteq \bigcup_{C\in\mathcal{C}} C.$$

Remark (Dependencies)*.*

- Subset
- Union

Definition (Subcover)

Definition 1.8.2 (Subcover)**.** $\mathcal{C}'$ is a *subcover* of $\mathcal{C}$ if $\mathcal{C}' \subseteq \mathcal{C}$ and $\mathcal{C}'$ is still a cover of A.

Remark (Dependencies)*.*

- Cover
- Subset

Definition (Finite Cover)

Definition 1.8.3 (Finite Cover)**.** A cover is *finite* if it has finitely many members. A *finite subcover* is a subcover $\mathcal{C}' \subseteq \mathcal{C}$ with $|\mathcal{C}'| < \infty$.

Remark (Dependencies).

- Cover
- Finite and Infinite Sets
- Subset

Remark (The key question). The condition of being a cover is easy to satisfy: take $\mathcal{C} = \mathcal{P}(A)$ and it is trivially a cover. The interesting question is whether large covers can be replaced by smaller ones — specifically, finite ones. An infinite cover has infinitely many sets, which is hard to work with. A finite subcover is just as good for covering purposes, but is finite and therefore tractable. When a finite subcover always exists, no matter which cover one starts with, the space is called compact.

Definition (Open Cover)

Definition 1.8.4 (Open Cover)**.** In a topological space (X, τ), an *open cover* of a subset $A \subseteq X$ is a cover $\mathcal{C}$ of A all of whose members are open sets: $\mathcal{C} \subseteq \tau$.

Remark (Standard quantified statement). $\mathcal{C}$ is an open cover of $A \leftrightarrow (\forall C \in \mathcal{C},\ C \in \tau) \wedge (\forall x \in A,\ \exists C \in \mathcal{C},\ x \in C)$.

Remark (Dependencies).

- Cover
- Finite Intersection Property
- Complement
- Intersection
- Subset

Remark (Why open covers). The emphasis on open covers in compactness is because open sets are the sets one controls in a topological space. An open cover $\{U_\alpha\}$ of A means: every point of A has at least one U_α containing it. A finite subcover then gives a list of finitely many open sets whose union already contains all of A. Compactness says this reduction is always possible, which is a very strong structural property.

Definition (Finite Intersection Property)

Definition 1.8.5 (Finite Intersection Property)**.** A collection $\mathcal{F}$ of sets has the *finite intersection property* (FIP) if every finite subcollection has nonempty intersection: for all finite $\mathcal{F}' \subseteq \mathcal{F}$,

$$\bigcap_{F \in \mathcal{F}'} F \neq \varnothing.$$

Remark (Dependencies).

- Intersection
- Empty Set

- Subset
- Finite and Infinite Sets

Remark (How to read the FIP). The FIP is a consistency condition: no finite list of sets from $\mathcal{F}$ is mutually exclusive. It does *not* say the entire intersection $\bigcap_{F\in\mathcal{F}} F$ is nonempty — only that every *finite* sub-intersection is. The difference matters when $\mathcal{F}$ is infinite.

Example: In $\mathbb{R}$, the family $\mathcal{F} = \{(0, 1/n) : n \in \mathbb{N}\}$ has the FIP (any finite subcollection has nonempty intersection, since $(0, 1/n) \supseteq (0, 1/N)$ for the largest N in the subcollection). But $\bigcap_{n=1}^{\infty}(0, 1/n) = \varnothing$. Compact spaces are characterized by the property that the FIP implies nonempty total intersection.

Proposition (FIP–Cover Duality)

Proposition 1.8.6 (FIP–Cover Duality)**.** *Let X be a set, $A \subseteq X$, and $\{U_\alpha\}_{\alpha\in I}$ a family of subsets of X. Let $F_\alpha = U_\alpha^c = X \setminus U_\alpha$. Then:*

$$\{U_\alpha\} \text{ covers } A \iff \{F_\alpha \cap A\}_{\alpha\in I} \text{ does not have the FIP.}$$

More precisely: the family $\{U_\alpha\}$ does not *cover A if and only if the family of closed complements $\{F_\alpha\}$ has the FIP relative to A. Go to proof.*

Remark (Standard quantified statement). $A \subseteq \bigcup_{\alpha\in I} U_\alpha \leftrightarrow \neg(\forall \text{ finite } I_0 \subseteq I,\ \bigcap_{\alpha\in I_0} F_\alpha \cap A \neq \varnothing)$. Reading pointwise: $\forall x \in A,\ \exists \alpha \in I,\ x \in U_\alpha \leftrightarrow \{F_\alpha \cap A\}$ fails the FIP.

Remark (Dependencies).

- Cover
- Finite Intersection Property
- Complement
- Intersection
- Subset

Remark (Proof by De Morgan). The duality is a direct consequence of De Morgan's law for indexed families (Theorem 1.7.4). The family $\{U_\alpha\}$ fails to cover A means:

$$A \not\subseteq \bigcup_\alpha U_\alpha \iff A \cap \Big(\bigcup_\alpha U_\alpha\Big)^c \neq \varnothing \iff A \cap \bigcap_\alpha U_\alpha^c \neq \varnothing \iff A \cap \bigcap_\alpha F_\alpha \neq \varnothing.$$

Applying this to every finite subcollection gives the equivalence with the FIP.

Remark (Why this duality matters for compactness). Compactness has two equivalent formulations:

(i) Every open cover has a finite subcover.

(ii) Every family of closed sets with the FIP has nonempty intersection.

These formulations are equivalent by FIP–Cover Duality: take complements and apply De Morgan. Formulation (i) is the standard definition; formulation (ii) is often easier to apply when working with closed sets (e.g., nested compact sets). Both are purely set-theoretic conditions, and the equivalence is entirely a matter of set algebra. When compactness is encountered in topology and metric spaces, the only new content is the geometric meaning — the set-theoretic equivalence is established here.

Example (Nested closed sets and FIP). In $\mathbb{R}$, the family $\{[n, \infty) : n \in \mathbb{N}\}$ has the FIP: any finite subcollection $\{[n_1, \infty), \dots, [n_k, \infty)\}$ has intersection $[\max(n_i), \infty) \neq \varnothing$. But $\bigcap_{n=1}^{\infty}[n, \infty) = \varnothing$. This shows $\mathbb{R}$ is not compact (the family of closed sets has the FIP but empty total intersection). A compact space would not allow this failure.

Chapter 2

Relations

relations

Set Theory → **Relations** → Functions

Remark (Source orientation). This chapter follows Tarski's treatment of relations as objects that can be formed, compared, reversed, and specialized into functional and one-one relations Tarski [1941].

2.1 Relations — Quick Reference

Relations — Quick Reference

Concept	Meaning	Detail
Ordered pair	$(a, b) := \{\{a\}, \{a, b\}\}$; identity iff both coords equal	↓ Def
Cartesian product	$A \times B$: all ordered pairs from A and B	↓ Def
Relation	Any subset $R \subseteq A \times B$	↓ Def
Domain of R	Inputs that occur in R	↓ Def
Range of R	Outputs that occur in R	↓ Def
Converse	Reverse every ordered pair	↓ Def
Identity relation	Diagonal relation Δ_A	↓ Def
Relation composition	Compose through an intermediate witness	↓ Def

Relation Algebra and Classification — Quick Reference

Concept	Meaning	Detail
Null relation	$\varnothing$	↓ Def
Universal relation	$A \times B$	↓ Def
Relation inclusion	$R \subseteq S$ as set inclusion	↓ Def
Relation equality	Mutual relation inclusion	↓ Def
Union of relations	Pair belongs to at least one relation	↓ Def
Intersection	Pair belongs to both relations	↓ Def
Difference	Pair belongs to first relation but not second	↓ Def
Complement	Pairs in $A \times B$ not in R	↓ Def
Converse laws	Converse reverses composition and preserves Boolean operations	↓ Thm
Composition laws	Associative; distributes over unions, not all Boolean operations	↓ Thm
One-to-many	One input has two distinct outputs	↓ Def
Many-to-one	Two distinct inputs share one output	↓ Def
Many-to-many	One-to-many and many-to-one phenomena both occur	↓ Def
Ternary relation	Subset of $A \times B \times C$	↓ Def

2.2 Ordered Pairs and Relations

Remark (Geometric picture). A binary relation from A to B can be pictured as directed edges from the A-side to the B-side of a bipartite diagram. The converse reverses every edge, and composition follows two edges in series through an intermediate set.

Definition (Ordered Pair)

Definition 2.2.1 (Ordered Pair)**.** Let a and b be sets. The *ordered pair* (a, b) is defined (Kuratowski) as

$$(a, b) \;:=\; \{\{a\}, \{a, b\}\}.$$

Remark (Dependencies).

- Axiom of Pairing

Theorem (Uniqueness of Ordered Pairs)

Theorem 2.2.2 (Uniqueness of Ordered Pairs)**.** *For any sets a, b, c, d,*

$$(a, b) = (c, d) \iff (a = c \wedge b = d).$$

Go to proof.

Remark (Standard quantified statement). $\forall a, b, c, d : \; (a, b) = (c, d) \;\leftrightarrow\; (a = c \wedge b = d)$.

Remark (Dependencies).

- Ordered Pair
- Axiom of Pairing
- Axiom of Extensionality

Remark (Kuratowski encoding). The purpose of the Kuratowski definition is purely to guarantee the uniqueness theorem above. Once $(a, b) = (c, d) \iff a = c \land b = d$ is established, we may treat ordered pairs as a primitive with this property and forget the encoding.

Definition (Cartesian Product — as foundation for relations)

Definition 2.2.3 (Cartesian Product — as foundation for relations)**.** The *Cartesian product* of sets A and B is:

$$A \times B \;:=\; \{\, (a, b) \mid a \in A \text{ and } b \in B \,\}.$$

Remark (Dependencies).

- Ordered Pair
- Axiom of Pairing
- Axiom of Power Set
- Axiom Schema of Separation

Definition (Relation)

Definition 2.2.4 (Relation)**.** A *relation* from A to B is any subset $R \subseteq A \times B$. If $(a, b) \in R$ we write $a\,R\,b$ and say "a is related to b." When $A = B$ we say R is a *relation on* A.

Remark (Dependencies).

- Cartesian Product
- Subset

Remark (Relations as structured sets). Relations introduce no new foundational objects: they are simply sets of ordered pairs, constructed using operations already available. Properties of relations can therefore be studied with ordinary set-theoretic reasoning.

Definition (Domain of a Relation)

Definition 2.2.5 (Domain of a Relation)**.** Let $R \subseteq A \times B$ be a relation. The *domain* of R is

$$\mathrm{Dom}(R) \;:=\; \{\, a \in A \mid \exists b \in B,\ (a, b) \in R \,\}.$$

Remark (Dependencies).

- Relation
- Axiom Schema of Separation

Definition (Range of a Relation)

Definition 2.2.6 (Range of a Relation)**.** Let $R \subseteq A \times B$ be a relation. The *range* of R is

$$\mathrm{Ran}(R) \; := \; \{\, b \in B \mid \exists a \in A,\; (a, b) \in R \,\}.$$

Tarski also calls this class the *counter-domain* or *converse domain*.

Remark (Dependencies).

- Relation
- Axiom Schema of Separation

Remark (Domain and range)*.* The domain records the first coordinates that occur in a relation; the range records the second coordinates that occur.

Definition (Converse Relation)

Definition 2.2.7 (Converse Relation)**.** If $R \subseteq A \times B$, the *converse* of R is the relation $R^{-1} \subseteq B \times A$ defined by

$$(b, a) \in R^{-1} \iff (a, b) \in R.$$

Equivalently, $b\,R^{-1}\,a$ iff aRb.

Remark (Dependencies).

- Relation
- Ordered Pair

Definition (Identity Relation)

Definition 2.2.8 (Identity Relation)**.** For a set A, the *identity relation* on A is

$$\Delta_A \; := \; \{\, (a, a) \mid a \in A \,\}.$$

Remark (Dependencies).

- Relation
- Cartesian Product

Definition (Composition of Relations)

Definition 2.2.9 (Composition of Relations)**.** Let $R \subseteq A \times B$ and $S \subseteq B \times C$. The *composition* $S \circ R \subseteq A \times C$ is defined by

$$(a, c) \in S \circ R \iff \exists b \in B,\; (a, b) \in R \text{ and } (b, c) \in S.$$

Remark (Dependencies).

- Relation

- Cartesian Product

Theorem (Associativity of Relation Composition)

Theorem 2.2.10 (Associativity of Relation Composition). *Let $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$. Then*

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

Go to proof.

Remark (Standard quantified statement). For all $a \in A$ and $d \in D$, $(a, d) \in T \circ (S \circ R)$ iff there are $b \in B$ and $c \in C$ with $(a, b) \in R$, $(b, c) \in S$, and $(c, d) \in T$, which is exactly the membership condition for $(a, d) \in (T \circ S) \circ R$.

Remark (Dependencies).

- Composition of Relations
- Relation Equality

Theorem (Identity Laws for Relation Composition)

Theorem 2.2.11 (Identity Laws for Relation Composition). *Let $R \subseteq A \times B$. Then*

$$R \circ \Delta_A = R \quad \text{and} \quad \Delta_B \circ R = R.$$

Go to proof.

Remark (Dependencies).

- Identity Relation
- Composition of Relations

Definition (Null Relation)

Definition 2.2.12 (Null Relation). For sets A and B, the *null relation* from A to B is $\varnothing \subseteq A \times B$. No element of A is related to any element of B.

Remark (Dependencies).

- Relation
- Empty Set
- Cartesian Product

Definition (Universal Relation)

Definition 2.2.13 (Universal Relation). For sets A and B, the *universal relation* from A to B is $A \times B$ itself. Every element of A is related to every element of B.

Remark (Dependencies).

- Relation

- Cartesian Product

Remark (Extreme relations). The null relation and universal relation are the two extreme binary relations from A to B.

Definition (Relation Inclusion)

Definition 2.2.14 (Relation Inclusion)**.** Let $R, S \subseteq A \times B$. Relation inclusion is ordinary set inclusion:

$$R \subseteq S \iff \forall a \in A \forall b \in B, \; ((a, b) \in R \Rightarrow (a, b) \in S),$$

Remark (Dependencies).

- Relation
- Subset

Definition (Relation Equality)

Definition 2.2.15 (Relation Equality)**.** Let $R, S \subseteq A \times B$. Relation equality is ordinary set equality:

$$R = S \iff R \subseteq S \text{ and } S \subseteq R.$$

Remark (Dependencies).

- Relation
- Relation Inclusion
- Set Equality

Definition (Union of Relations)

Definition 2.2.16 (Union of Relations)**.** Let $R, S \subseteq A \times B$. The *union* $R \cup S$ is the relation with

$$(a, b) \in R \cup S \iff (a, b) \in R \text{ or } (a, b) \in S.$$

Remark (Dependencies).

- Relation
- Union

Definition (Intersection of Relations)

Definition 2.2.17 (Intersection of Relations)**.** Let $R, S \subseteq A \times B$. The *intersection* $R \cap S$ is the relation with

$$(a, b) \in R \cap S \iff (a, b) \in R \text{ and } (a, b) \in S.$$

Remark (Dependencies).

- Relation

- Intersection

Definition (Difference of Relations)

Definition 2.2.18 (Difference of Relations)**.** Let $R, S \subseteq A \times B$. The *difference* $R \setminus S$ is the relation with

$$(a, b) \in R \setminus S \iff (a, b) \in R \text{ and } (a, b) \notin S.$$

Remark (Dependencies).

- Relation
- Set Difference

Definition (Complement of a Relation)

Definition 2.2.19 (Complement of a Relation)**.** Let $R \subseteq A \times B$. The *complement* of R relative to $A \times B$ is

$$(A \times B) \setminus R.$$

Remark (Dependencies).

- Relation
- Set Difference

Remark (Relation algebra operations). Relation inclusion, equality, union, intersection, difference, and complement are inherited from the corresponding set operations on subsets of $A \times B$.

Theorem (Converse Laws for Relation Operations)

Theorem 2.2.20 (Converse Laws for Relation Operations)**.** *Let $R, S \subseteq A \times B$, let $T \subseteq B \times C$, and take complements relative to the indicated Cartesian products. Then*

$$(R^{-1})^{-1} = R, \qquad (T \circ R)^{-1} = R^{-1} \circ T^{-1},$$

$$(R \cup S)^{-1} = R^{-1} \cup S^{-1}, \qquad (R \cap S)^{-1} = R^{-1} \cap S^{-1},$$
$$(R \setminus S)^{-1} = R^{-1} \setminus S^{-1}, \qquad ((A \times B) \setminus R)^{-1} = (B \times A) \setminus R^{-1}.$$

Go to proof.

Remark (Dependencies).

- Converse Relation
- Composition of Relations
- Union of Relations
- Intersection of Relations
- Difference of Relations

- Complement of a Relation

Theorem (Relation Composition and Boolean Operations)

Theorem 2.2.21 (Relation Composition and Boolean Operations)**.** *Let $P, Q \subseteq A \times B$, $R \subseteq B \times C$, and $S, T \subseteq B \times C$. Then*

$$R \circ (P \cup Q) = (R \circ P) \cup (R \circ Q), \qquad (S \cup T) \circ P = (S \circ P) \cup (T \circ P).$$

Also,

$$R \circ (P \cap Q) \subseteq (R \circ P) \cap (R \circ Q), \qquad (S \cap T) \circ P \subseteq (S \circ P) \cap (T \circ P).$$

The analogous formulas for difference and complement hold only as inclusions or under extra hypotheses; they are not general distributive laws. *Go to proof.*

Remark (Dependencies).

- Composition of Relations
- Union of Relations
- Intersection of Relations

Remark (Warning on relation composition). Relation composition distributes over unions on both sides. It does not generally distribute over intersections, differences, or complements. The reason is existential: a pair can lie in both composed relations using two different middle witnesses, even when no single witness satisfies both conditions at once.

Remark (Relation algebra interaction table).

Operation	**Converse**	**Composition**
Union	$(R \cup S)^{-1} = R^{-1} \cup S^{-1}$	Distributes exactly on both sides
Intersection	$(R \cap S)^{-1} = R^{-1} \cap S^{-1}$	Gives $\subseteq$ in general; equality needs extra hypotheses
Difference	$(R \setminus S)^{-1} = R^{-1} \setminus S^{-1}$	No general distributive equality
Complement	$((A \times B) \setminus R)^{-1} = (B \times A) \setminus R^{-1}$	No general distributive equality
Composition	$(T \circ R)^{-1} = R^{-1} \circ T^{-1}$	Associative with identity relation Δ_A

Definition (Transitive Closure)

Definition 2.2.22 (Transitive Closure)**.** Let R be a relation on A. The *transitive closure* of R is the smallest transitive relation on A containing R.

Remark (Dependencies).

- Relation
- Transitive Relation
- Relation Inclusion

Definition (Equivalence Closure)

Definition 2.2.23 (Equivalence Closure)**.** Let R be a relation on A. The *equivalence closure* of R is the smallest equivalence relation on A containing R.

Remark (Dependencies).

- Relation
- Equivalence Relation
- Relation Inclusion

Definition (One-to-Many Relation)

Definition 2.2.24 (One-to-Many Relation)**.** Let $R \subseteq A \times B$. R is *one-to-many* if some input is related to two distinct outputs:

$$\exists a \in A \, \exists b_1, b_2 \in B, \quad (a, b_1) \in R \wedge (a, b_2) \in R \wedge b_1 \neq b_2.$$

Remark (Dependencies).

- Relation

Definition (Many-to-One Relation)

Definition 2.2.25 (Many-to-One Relation)**.** Let $R \subseteq A \times B$. R is *many-to-one* if two distinct inputs are related to a common output:

$$\exists a_1, a_2 \in A \, \exists b \in B, \quad (a_1, b) \in R \wedge (a_2, b) \in R \wedge a_1 \neq a_2.$$

Remark (Dependencies).

- Relation

Definition (Many-to-Many Relation)

Definition 2.2.26 (Many-to-Many Relation)**.** Let $R \subseteq A \times B$. R is *many-to-many* if it is one-to-many and many-to-one.

Remark (Dependencies).

- One-to-Many Relation
- Many-to-One Relation

Remark (Relation cardinality patterns). The one-to-many, many-to-one, and many-to-many labels classify how many inputs and outputs can be connected by a relation.

Remark (Why one-to-many is not functional). A one-to-many relation fails the determinacy condition needed for a function: one input admits at least two possible outputs. A many-to-one relation may still be a function, because different inputs are allowed to share an output.

Definition (Many-Place Relation)

Definition 2.2.27 (Many-Place Relation)**.** A *ternary relation* among sets A, B, and C is a subset of $A \times B \times C$. More generally, an n-place relation on $A_1, \dots, A_n$ is a subset of

$$A_1 \times \cdots \times A_n.$$

For example, betweenness in geometry is naturally a three-place relation, and the arithmetic statement $x + y = z$ is naturally a three-place relation among x, y, and z.

Remark (Dependencies).

- Relation
- Cartesian Product

Remark (Tarski-style hierarchy). The hierarchy used in the next chapter is:

$$\begin{aligned} \text{relation} &\longrightarrow \text{functional relation} \longrightarrow \text{function} \\ &\longrightarrow \text{one-one or onto function} \longrightarrow \text{bijection}. \end{aligned}$$

The first step adds determinacy; the second adds totality; the later steps control whether outputs are shared or missed.

Relational Properties — Quick Reference

Property	**Formal definition**	**Canonical example**	**Detail**
Reflexive	$\forall a,\ (a,a) \in R$	$=$ on any set	↓ Def
Irreflexive	$\forall a,\ (a,a) \notin R$	$<$ on $\mathbb{N}$	↓ Def
Symmetric	$(a,b) \in R \Rightarrow (b,a) \in R$	"same age as"	↓ Def
Antisymmetric	$(a,b),(b,a) \in R \Rightarrow a = b$	$\leq$ on $\mathbb{N}$	↓ Def
Asymmetric	$(a,b) \in R \Rightarrow (b,a) \notin R$	$<$ on $\mathbb{N}$	↓ Def
Transitive	$(a,b),(b,c) \in R \Rightarrow (a,c) \in R$	$\leq$ on $\mathbb{N}$	↓ Def
Total (Connex)	$(a,b) \in R \vee (b,a) \in R$	$\leq$ on $\mathbb{R}$	↓ Def
Structural classes formed by combining properties:			
Equivalence	Reflexive + Symmetric + Transitive	$=$ on any set	↓ Def
Preorder	Reflexive + Transitive	$\leq$ on $\mathbb{N}$	↓ Def
Partial order	Reflexive + Antisymmetric + Transitive	$\subseteq$ on $\mathcal{P}(A)$	↓ Def
Total order	Partial order + Total	$\leq$ on $\mathbb{R}$	↓ Def

2.3 Properties of Relations

Definition (Reflexive Relation)

Definition 2.3.1 (Reflexive Relation)**.** A binary relation $R \subseteq A \times A$ is *reflexive* on A if every element of the domain is related to itself:

$$\forall a \in A,\ (a,a) \in R.$$

In infix notation, this is the condition $\forall a \in A,\ aRa$.

Remark (Standard quantified statement). Reflexivity is a universal requirement over the whole domain:

$$\mathrm{Reflexive}_A(R) \iff \forall a \in A,\ (a,a) \in R.$$

For first-order logic with equality, the corresponding logical baseline is the identity axiom

$$\forall x\,(x = x),$$

which forces equality to behave reflexively in every structure.

Remark (Dependencies).

- Relation
- Universal Formula

Definition (Irreflexive Relation)

Definition 2.3.2 (Irreflexive Relation)**.** A binary relation $R \subseteq A \times A$ is *irreflexive* on A if no element of the domain is related to itself:

$$\forall a \in A,\ (a,a) \notin R.$$

In infix notation, this is the condition $\forall a \in A,\ \neg(aRa)$.

Remark (Standard quantified statement). Irreflexivity is not the same as merely failing to be reflexive:

$$\mathrm{Irreflexive}_A(R) \iff \forall a \in A,\ \neg(aRa),$$

whereas

$$\neg\,\mathrm{Reflexive}_A(R) \iff \exists a \in A,\ \neg(aRa).$$

Thus a relation may be neither reflexive nor irreflexive. In first-order axiomatizations of strict order, irreflexivity is commonly imposed by the axiom

$$\forall x\,\neg(x < x).$$

Remark (Dependencies).

- Relation
- Universal Formula

Remark (Independence of reflexive and irreflexive). A relation may be neither reflexive nor irreflexive (if some but not all elements satisfy $(a,a) \in R$). However, a relation cannot be both reflexive and irreflexive unless $A = \varnothing$.

Definition (Symmetric Relation)

Definition 2.3.3 (Symmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *symmetric* on A if every related pair also appears in the reverse order:

$$\forall a,b \in A,\ (a,b) \in R \to (b,a) \in R.$$

In infix notation, this is $\forall a,b \in A,\ aRb \to bRa$.

Remark (Standard quantified statement). Symmetry is a conditional property:

$$\text{Symmetric}_A(R) \iff \forall a, b \in A,\ (aRb \to bRa).$$

It does not require every pair to be related; it requires only that any existing directed relation can be reversed. For first-order logic with equality, symmetry is expressed by the identity axiom

$$\forall x \forall y\, (x = y \to y = x).$$

Remark (Dependencies).

- Relation
- Universal Formula

Definition (Antisymmetric Relation)

Definition 2.3.4 (Antisymmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *antisymmetric* on A if any two elements related in both directions must be equal:

$$\forall a, b \in A,\ \big((a, b) \in R \land (b, a) \in R\big) \to a = b.$$

In infix notation, this is $\forall a, b \in A,\ ((aRb \land bRa) \to a = b)$.

Remark (Standard quantified statement). Antisymmetry forbids two-way relation cycles between distinct elements:

$$\text{Antisymmetric}_A(R) \iff \forall a, b \in A,\ ((aRb \land bRa) \to a = b).$$

It is not the same as asymmetry. Antisymmetry permits aRa; asymmetry forbids it. Thus $\leq$ is antisymmetric, while $<$ is asymmetric. In first-order axiomatizations of partial orders, antisymmetry is usually written

$$\forall x \forall y\, ((x \leq y \land y \leq x) \to x = y).$$

Remark (Dependencies).

- Relation
- Universal Formula

Definition (Asymmetric Relation)

Definition 2.3.5 (Asymmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *asymmetric* on A if every related pair excludes the reverse pair:

$$\forall a, b \in A,\ (a, b) \in R \to (b, a) \notin R.$$

In infix notation, this is $\forall a, b \in A,\ (aRb \to \neg(bRa))$.

Remark (Standard quantified statement). Asymmetry is the strict one-way condition

$$\text{Asymmetric}_A(R) \iff \forall a, b \in A,\ (aRb \to \neg(bRa)).$$

It implies irreflexivity by taking $a = b$, and it implies antisymmetry because two-way relation between distinct elements is impossible. Conversely, a relation that is both irreflexive and antisymmetric is asymmetric. For strict orders, the typical first-order axiom is

$$\forall x \forall y\, (x < y \to \neg(y < x)).$$

Remark (Dependencies).

- Relation
- Universal Formula

Remark (Asymmetric vs. antisymmetric). Every asymmetric relation is antisymmetric, but not conversely. $\leq$ on $\mathbb{R}$ is antisymmetric but not asymmetric (since $1 \leq 1$). Strict order $<$ is both asymmetric and irreflexive.

Definition (Transitive Relation)

Definition 2.3.6 (Transitive Relation)**.** A binary relation $R \subseteq A \times A$ is *transitive* on A if any two composable related pairs collapse to a direct related pair:

$$\forall a, b, c \in A,\ \big((a,b) \in R \wedge (b,c) \in R\big) \to (a,c) \in R.$$

In infix notation, this is $\forall a, b, c \in A,\ ((aRb \wedge bRc) \to aRc)$.

Remark (Standard quantified statement). Transitivity is the formal chaining law:

$$\text{Transitive}_A(R) \iff \forall a, b, c \in A,\ ((aRb \wedge bRc) \to aRc).$$

For equality, the corresponding first-order axiom is

$$\forall x \forall y \forall z\, ((x = y \wedge y = z) \to x = z).$$

Together with reflexivity and symmetry, transitivity gives equivalence relations; together with reflexivity and antisymmetry, it gives partial orders.

Remark (Dependencies).

- Relation
- Universal Formula

Definition (Total (Connex) Relation)

Definition 2.3.7 (Total (Connex) Relation)**.** R is *total* (or *connex*) if every pair is comparable:

$$\forall a, b \in A,\ (a,b) \in R \vee (b,a) \in R.$$

Remark (Dependencies).

- Relation
- Universal Formula

Definition (Equivalence Relation)

Definition 2.3.8 (Equivalence Relation)**.** R is an *equivalence relation* on A if it is reflexive, symmetric, and transitive.

Remark (Standard quantified statement)*.* R is an equivalence relation on A iff:

$$(\forall a \in A,\ (a,a) \in R) \wedge (\forall a,b \in A,\ (a,b) \in R \to (b,a) \in R) \wedge (\forall a,b,c \in A,\ (a,b) \in R \wedge (b,c) \in R \to (a,$$

Remark (Role in mathematics)*.* Equivalence relations axiomatize the idea of "sameness up to a chosen criterion." They partition sets into equivalence classes and are the basis for quotient constructions throughout algebra, topology, and analysis.

Definition (Preorder)

Definition 2.3.9 (Preorder)**.** R is a *preorder* on A if it is reflexive and transitive.

Remark (Standard quantified statement)*.* R is a preorder on A iff:

$$(\forall a \in A,\ (a,a) \in R)\ \wedge\ (\forall a,b,c \in A,\ (a,b) \in R \wedge (b,c) \in R \to (a,c) \in R).$$

Remark (Dependencies)*.*

- Reflexive Relation
- Transitive Relation

Definition (Partial Order)

Definition 2.3.10 (Partial Order)**.** R is a *partial order* on A if it is reflexive, antisymmetric, and transitive.

Remark (Standard quantified statement)*.* R is a partial order on A iff:

$$(\forall a \in A,\ (a,a) \in R) \wedge (\forall a,b \in A,\ (a,b) \in R \wedge (b,a) \in R \to a = b) \wedge (\forall a,b,c \in A,\ (a,b) \in R \wedge (b,c) \in R$$

Remark (Dependencies)*.*

- Reflexive Relation
- Antisymmetric Relation
- Transitive Relation

Definition (Total Order)

Definition 2.3.11 (Total Order)**.** R is a *total order* on A if it is a partial order and is total.

Remark (Standard quantified statement). R is a total order on A iff R is a partial order on A and additionally:

$$\forall a, b \in A,\ (a, b) \in R \vee (b, a) \in R.$$

Remark (Dependencies).

- Partial Order
- Total Relation

Remark (Using structural properties in proofs). When R is asserted to be an equivalence relation or partial order, this is shorthand for the conjunction of its defining properties. In proofs, cite only the specific component needed: "since R is a partial order, antisymmetry implies ..." rather than restating all properties.

Example (Using a structural property in a proof). Let R be an equivalence relation on A. For any $a, b \in A$, $(a, b) \in R \Rightarrow [a] = [b]$, where $[a] = \{x \in A \mid (a, x) \in R\}$.

Remark (Properties are logically independent). No basic property implies another in general. A relation may be transitive without being reflexive, or symmetric without being transitive. The structural classes are distinguished precisely by requiring specific combinations.

2.4 Equivalence

Equivalence and Partitions — Quick Reference

Concept	**Meaning**	**Detail**
Equivalence class	$[a]_R = \{x \in A \mid (a, x) \in R\}$	↓ Def
Quotient set	A/R: all equivalence classes	↓ Def
Index of R	Cardinality of A/R	↓ Def
Canonical surjection	$\pi : A \to A/R$, $\pi(a) = [a]$	↓ Def
Partition	Nonempty, disjoint, covering collection of subsets	↓ Def
Rep. independence	$[a] = [b] \iff (a, b) \in R$	↓ Lem
Equiv.–partition correspondence	Bijection between equiv. relations and partitions	↓ Thm

2.5 Equivalence Classes and Partitions

Definition (Equivalence Class)

Definition 2.5.1 (Equivalence Class)**.** Let R be an equivalence relation on A. For $a \in A$, the *equivalence class* of a is:

$$[a]_R := \{x \in A \mid (a, x) \in R\}.$$

When R is clear from context, we write $[a]$.

Remark (Dependencies).

- Equivalence Relation

- Relation

Remark (English reading). $[a]_R$ is the set of all elements that R declares "the same as a." Two elements lie in the same class iff they are related: this is precisely the Representative Independence Lemma below.

Definition (Quotient Set)

Definition 2.5.2 (Quotient Set)**.** The *quotient set* of A by R is the set of all equivalence classes:

$$A/R \;:=\; \{\, [a]_R \mid a \in A \,\}.$$

Remark (Dependencies).

- Equivalence Relation
- Equivalence Class

Definition (Index of an Equivalence Relation)

Definition 2.5.3 (Index of an Equivalence Relation)**.** The *index* of R on A is the cardinality $|A/R|$, i.e. the number of equivalence classes.

Remark (Standard quantified statement). $\mathrm{index}(R) = |A/R| = |\{[a]_R \mid a \in A\}|$.

Remark (Dependencies).

- Quotient Set
- Same Cardinality

Definition (Canonical Surjection)

Definition 2.5.4 (Canonical Surjection)**.** The *canonical surjection* (quotient map) is the function

$$\pi : A \to A/R, \qquad \pi(a) := [a].$$

Remark (Forward reference). This definition uses the function vocabulary developed in the next chapter. It is included here because quotient sets are naturally introduced with their canonical map; proofs using its function-theoretic properties may be revisited after the function chapter.

Remark (Dependencies).

- Function
- Quotient Set
- Equivalence Class
- Surjective Function

Remark (Properties of π). π is surjective by construction. Elements $a, b \in A$ satisfy $\pi(a) = \pi(b)$ iff $(a, b) \in R$. The canonical surjection is the prototype for all quotient constructions: it reappears as the quotient homomorphism in algebra and the quotient map in topology.

Definition (Partition)

Definition 2.5.5 (Partition)**.** A *partition* of A is a collection $\mathcal{P}$ of subsets of A such that:

(i) every block is nonempty: $\forall P \in \mathcal{P},\ P \neq \varnothing$;

(ii) distinct blocks are disjoint: $\forall P, Q \in \mathcal{P},\ P \neq Q \to P \cap Q = \varnothing$;

(iii) the blocks cover A: $\bigcup_{P \in \mathcal{P}} P = A$.

The sets $P \in \mathcal{P}$ are called the *blocks* of the partition.

Remark (Dependencies).

- Subset
- Empty Set
- Intersection
- Union
- Cover

Remark (Partition vs. cover). Every partition of A is a cover of A whose members are nonempty and pairwise disjoint. Partitions are precisely the covers satisfying the disjointness condition.

Lemma (Representative Independence Lemma)

Lemma 2.5.6 (Representative Independence Lemma)**.** *Let R be an equivalence relation on A. For any $a, b \in A$,*

$$[a] = [b] \iff (a, b) \in R.$$

Go to proof.

Remark (Standard quantified statement). $\forall a, b \in A : \ [a]_R = [b]_R \leftrightarrow (a, b) \in R.$

Remark (Dependencies).

- Equivalence Relation
- Equivalence Class
- Partition
- Quotient Set

Remark (Well-definedness on quotient sets)*.* This lemma is the key fact underlying well-definedness of functions on quotient sets: a function $f : A \to B$ defined by $f([a]) = \cdots$ is well-defined iff the formula gives the same output for all representatives of $[a]$.

Theorem (Equivalence Relations and Partitions)

Theorem 2.5.7 (Equivalence Relations and Partitions)**.** *Let A be a set.*

(i) If R is an equivalence relation on A, then A/R is a partition of A.

(ii) If $\mathcal{P}$ is a partition of A, then the relation $R_{\mathcal{P}}$ defined by

$$(a, b) \in R_{\mathcal{P}} \iff \exists P \in \mathcal{P} \text{ with } a \in P \text{ and } b \in P$$

is an equivalence relation on A.

(iii) These constructions are inverse: $R_{A/R} = R$ and $A/R_{\mathcal{P}} = \mathcal{P}$.

Go to proof.

Remark (Standard quantified statement)*.* (i) $\forall a \in A,\ [a]_R \neq \varnothing$; $\forall a, b \in A,\ [a]_R \neq [b]_R \to [a]_R \cap [b]_R = \varnothing$; $\bigcup_{a \in A} [a]_R = A$.
(ii) $\forall a, b \in A :\ (a, b) \in R_{\mathcal{P}} \leftrightarrow \exists P \in \mathcal{P},\ a \in P \wedge b \in P$.

Remark (Dependencies)*.*

- Equivalence Relation
- Equivalence Class
- Quotient Set
- Canonical Surjection
- Function
- Composition

Remark (Duality of perspectives)*.* This theorem establishes a bijection between equivalence relations on A and partitions of A. The two perspectives—"same block" (partition) and "related" (equivalence relation)—are interchangeable and each is more natural in different contexts.

Example (Extremal equivalence relations)*.* On any set A:

1. The *equality relation* $(a, b) \in R \iff a = b$ gives singleton classes $[a] = \{a\}$ and the finest partition of A.
2. The *universal relation* $R = A \times A$ gives $[a] = A$ for all a, and the coarsest partition (one block).

All other equivalence relations on A lie strictly between these extremes.

Theorem (Universal Property of the Quotient Map)

Theorem 2.5.8 (Universal Property of the Quotient Map)**.** *Let R be an equivalence relation on A, let $\pi : A \to A/R$ be the canonical surjection, and let $f : A \to B$ be any function. Then f is* constant on equivalence classes *— meaning $(a, b) \in R$ implies $f(a) = f(b)$ — if and only if there exists a unique function $\bar{f} : A/R \to B$ such that*

$$f = \bar{f} \circ \pi.$$

When it exists, $\bar{f}$ is defined by $\bar{f}([a]) = f(a)$. Go to proof.

Remark (Forward reference)*.* This universal property depends on functions, surjectivity, and composition. It is stated here to complete the quotient-set picture, but its dependency ceiling is intentionally forward-looking.

Remark (Standard quantified statement)*.* $\big(\forall a, b \in A,\ (a, b) \in R \to f(a) = f(b)\big) \leftrightarrow \big(\exists!\, \bar{f} : A/R \to B,\ \forall a \in A,\ \bar{f}([a]) = f(a)\big)$.

Remark (Dependencies)*.*

- Equivalence Relation
- Canonical Surjection
- Quotient Set
- Function
- Composition

Remark (Why this is called the universal property)*.* The condition $f = \bar{f} \circ \pi$ says the diagram commutes: going from A to B directly via f gives the same result as going first to A/R via π and then to B via $\bar{f}$. The word "universal" refers to the fact that $\pi : A \to A/R$ is the *unique* quotient with this property: any other map that "respects the equivalence" factors through π.

The uniqueness of $\bar{f}$ is forced by π being surjective: every element of A/R is $[a]$ for some a, so $\bar{f}([a]) = f(a)$ is the only possible definition.

Remark (Well-definedness is the key verification)*.* The heart of the theorem is well-definedness: the formula $\bar{f}([a]) = f(a)$ could be ambiguous if $[a] = [b]$ but $f(a) \neq f(b)$. The hypothesis that f is constant on equivalence classes is precisely what prevents this: $(a, b) \in R$ (i.e., $[a] = [b]$) implies $f(a) = f(b)$, so the output does not depend on which representative of the class is chosen.

This "well-definedness check" recurs throughout mathematics. Whenever a function is defined on equivalence classes by a formula involving representatives, the first obligation is to verify that the formula gives the same answer for all representatives. The universal property tells exactly what condition on f makes this work.

Remark (Where this appears downstream)*.* This theorem is not an abstract curiosity. It is the engine behind:

- *Integer construction*: the integers $\mathbb{Z}$ are defined as equivalence classes of pairs of natural numbers, and arithmetic operations are defined by formulas on representatives. The universal property verifies the operations are well-defined.

- *Rational construction*: $\mathbb{Q}$ is defined as equivalence classes of pairs (p, q) with $q \neq 0$, and addition/multiplication on representatives must be checked.
- *Quotient groups in algebra*: the quotient group G/N is defined by the same factorization, and the quotient homomorphism is π.
- *Quotient topology*: the quotient topological space is defined as the finest topology on A/R making π continuous — a condition stated precisely via the universal property.

Chapter 3

Functions

functions

Relations → **Functions** → Orderings

Remark (Source orientation). This chapter follows Tarski's treatment of functions as special relations: first many-one relations, then one-one relations and bijective correspondences Tarski [1941].

3.1 Functions — Quick Reference

Functions — Quick Reference

Concept	Meaning	Detail
Functional relation	Each input has at most one output	↓ Def
Total relation on A	Each input has at least one output	↓ Def
Function	Left-total, right-unique relation	↓ Def
Partial function	Functional, not necessarily total	↓ Def
Function equality	Same domain, codomain, and pointwise values	↓ Def
Many-place function	Functional relation in the last coordinate	↓ Def
Domain	$\mathrm{dom}(f) = A$ for $f : A \to B$	↓ Def
Codomain	$\mathrm{cod}(f) = B$ for $f : A \to B$	↓ Def
Image of function	$\mathrm{im}(f) = \{f(a) \mid a \in A\}$	↓ Def
Image of set	$f(S) = \{f(a) \mid a \in S\}$ for $S \subseteq A$	↓ Def
Preimage	$f^{-1}(T) = \{a \in A \mid f(a) \in T\}$ for $T \subseteq B$	↓ Def
Fiber	$f^{-1}(\{b\})$: preimage of a singleton	↓ Def
Graph	$\{(a, b) \in A \times B \mid b = f(a)\}$	↓ Def
Injective	Distinct inputs $\Rightarrow$ distinct outputs	↓ Def
Surjective	Every codomain element is achieved	↓ Def
Bijective	Injective and surjective	↓ Def
Function test	Function iff total and functional	↓ Thm
One-one test	Injective iff converse graph is functional	↓ Prop
Bijective test	Unique correspondence in both directions	↓ Prop
Identity	$\mathrm{id}_A(a) = a$	↓ Def
Inclusion map	$\iota : A \hookrightarrow B$, $\iota(a) = a$ for $A \subseteq B$	↓ Def
Composition	$(g \circ f)(a) = g(f(a))$	↓ Def
Inverse	Defined for bijections; $f^{-1} \circ f = \mathrm{id}_A$	↓ Def
Left inverse	$g \circ f = \mathrm{id}_A$	↓ Def
Right inverse	$f \circ h = \mathrm{id}_B$	↓ Def
Restriction	$f\|_C : C \to B$	↓ Def
Extension	$g : A' \to B$ with $g\|_A = f$	↓ Def
Empty function	Unique function $\varnothing \to B$	↓ Def

3.2 Functions

Remark (Geometric picture). A function is a deterministic machine: every input in the domain is accepted, and no input can fork into two different outputs. Fibers are the level sets sitting over points of the codomain; they collect all inputs that the machine sends to the same output.

Definition (Functional Relation)

Definition 3.2.1 (Functional Relation)**.** Let $R \subseteq A \times B$ be a relation. We say that R is *functional from A to B* if each input has at most one output:

$$\forall a \in A\, \forall b_1, b_2 \in B, \quad \big((a, b_1) \in R \wedge (a, b_2) \in R\big) \Rightarrow b_1 = b_2.$$

Remark (Dependencies).

- Relation
- Cartesian Product

Definition (Total Relation on a Domain)

Definition 3.2.2 (Total Relation on a Domain)**.** Let $R \subseteq A \times B$ be a relation. We say that R is *total on* A if each input has at least one output:

$$\forall a \in A\, \exists b \in B, \quad (a, b) \in R.$$

Remark (Dependencies).

- Relation

Definition (Function)

Definition 3.2.3 (Function)**.** Let A and B be sets. A *function* from A to B is a relation $f \subseteq A \times B$, together with the declared domain A and codomain B, such that:

(i) (*Existence / left-total*) for every $a \in A$, there exists $b \in B$ with $(a, b) \in f$;

(ii) (*Uniqueness / right-unique*) if $(a, b_1) \in f$ and $(a, b_2) \in f$ then $b_1 = b_2$.

If $(a, b) \in f$ we write $f(a) = b$.

Remark (Dependencies).

- Relation
- Cartesian Product

Remark (English reading). A function is a rule assigning to each input exactly one output. The two conditions formalize "defined everywhere" (existence) and "single-valued" (uniqueness).

Remark (Function as relation). Every function is a relation, but not every relation is a function. The two conditions that distinguish functions are left-totality and right-uniqueness. The codomain is part of the function data, not something recoverable from the graph alone.

Theorem (Functional Relation Characterization of Functions)

Theorem 3.2.4 (Functional Relation Characterization of Functions)**.** *Let A and B be sets, and let $R \subseteq A \times B$. Then R is a function from A to B if and only if*

$$\forall a \in A\, \exists! b \in B, \quad (a, b) \in R.$$

Equivalently, R is a function from A to B if and only if R is total on A and functional from A to B. Go to proof.

Remark (Dependencies).

- Function

- Functional Relation
- Total Relation on a Domain

Definition (Partial Function)

Definition 3.2.5 (Partial Function)**.** Let A and B be sets. A *partial function* from A to B is a functional relation $R \subseteq A \times B$. It need not be total on A: some inputs may have no output.

Remark (Dependencies).

- Functional Relation

Proposition (Partial Functions Are Functional Relations)

Proposition 3.2.6 (Partial Functions Are Functional Relations)**.** *For sets A and B and a relation $R \subseteq A \times B$, R is a partial function from A to B if and only if R is functional from A to B.* *Go to proof.*

Remark (Dependencies).

- Partial Function
- Functional Relation

Definition (Function Equality)

Definition 3.2.7 (Function Equality)**.** For functions $f : A \to B$ and $g : C \to D$, we write $f = g$ if

$$A = C, \qquad B = D, \qquad \text{and} \qquad \forall a \in A,\ f(a) = g(a).$$

Remark (Dependencies).

- Function
- Set Equality

Proposition (One-to-Many Relations Are Not Functional)

Proposition 3.2.8 (One-to-Many Relations Are Not Functional)**.** *If $R \subseteq A \times B$ is one-to-many, then R is not functional from A to B. Consequently, R is not a function from A to B.* *Go to proof.*

Remark (Dependencies).

- One-to-Many Relation
- Functional Relation
- Function

Remark (Many-to-one is allowed). A function may send different inputs to the same output. That is the ordinary many-one case in Tarski's terminology. What is forbidden for a function is not shared output, but one input having two distinct outputs.

Definition (Domain of a Function)

Definition 3.2.9 (Domain of a Function)**.** If f is a function from A to B, we write $f : A \to B$, where A is the *domain* $\mathrm{dom}(f)$.

Remark (Dependencies).

- Function

Definition (Codomain of a Function)

Definition 3.2.10 (Codomain of a Function)**.** If f is a function from A to B, we write $f : A \to B$, where B is the *codomain* $\mathrm{cod}(f)$.

Remark (Dependencies).

- Function

Remark (Domain and codomain). The notation $f : A \to B$ records both the domain A and the codomain B. The same graph may define maps into larger codomains, but surjectivity and bijectivity depend on the declared codomain.

Definition (Image of a Function)

Definition 3.2.11 (Image of a Function)**.** The *image* (or *range*) of $f : A \to B$ is:

$$\mathrm{im}(f) \ := \ \{ \, b \in B \mid \exists a \in A, \ f(a) = b \, \}.$$

The image may be a proper subset of the codomain.

Remark (Dependencies).

- Function
- Set and Membership
- Axiom Schema of Separation

Remark (Range and image). For a function viewed as its graph, the relation range and the function image coincide. The codomain may be larger than this range/image.

Definition (Image of a Set)

Definition 3.2.12 (Image of a Set)**.** For $S \subseteq A$, the *image of S under f* is:

$$f(S) \ := \ \{ \, f(a) \mid a \in S \, \}.$$

Note $f(A) = \mathrm{im}(f)$.

Remark (Dependencies).

- Function
- Subset
- Axiom Schema of Separation

Definition (Preimage)

Definition 3.2.13 (Preimage)**.** For $T \subseteq B$, the *preimage* (inverse image) of T under f is:

$$f^{-1}(T) \ := \ \{\, a \in A \mid f(a) \in T \,\}.$$

Remark (Dependencies).

- Function
- Subset
- Axiom Schema of Separation

Remark (Preimage notation warning). $f^{-1}(T)$ does *not* denote an inverse function. It is defined for any function f, regardless of whether f is injective or bijective.

Definition (Fiber)

Definition 3.2.14 (Fiber)**.** The *fiber* of f over $b \in B$ is the preimage of the singleton:

$$f^{-1}(\{b\}) \ = \ \{\, a \in A \mid f(a) = b \,\}.$$

Remark (Dependencies).

- Function
- Preimage

Remark (Fibers partition the domain). The collection $\{f^{-1}(\{b\}) \mid b \in \mathrm{im}(f)\}$ is a partition of A. Every function thus induces an equivalence relation $a_1 \sim_f a_2 \iff f(a_1) = f(a_2)$, whose equivalence classes are precisely the fibers.

Definition (Graph of a Function)

Definition 3.2.15 (Graph of a Function)**.** The *graph* of $f : A \to B$ is:

$$\mathrm{Graph}(f) \ := \ \{\, (a, b) \in A \times B \mid b = f(a) \,\}.$$

Remark (Dependencies).

- Function
- Ordered Pair
- Cartesian Product

Remark (Function as graph). A relation $G \subseteq A \times B$ is the graph of a function $f : A \to B$ iff G is left-total and right-unique. Functions may therefore be identified with their graphs only after the domain and codomain have been fixed. As bare sets of ordered pairs, graphs do not encode the codomain.

Definition (Injective Function)

Definition 3.2.16 (Injective Function)**.** $f : A \to B$ is *injective* (one-to-one) if distinct elements of A have distinct images:

$$\forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \implies a_1 = a_2.$$

Remark (Dependencies).

- Function

Definition (Empty Function)

Definition 3.2.17 (Empty Function)**.** For every set B, the *empty function* from $\varnothing$ to B is the empty relation

$$\varnothing : \varnothing \to B.$$

It is the unique function with domain $\varnothing$ and codomain B.

Remark (Dependencies).

- Function
- Empty Set
- Codomain of a Function

Definition (Surjective Function)

Definition 3.2.18 (Surjective Function)**.** $f : A \to B$ is *surjective* (onto) if every element of B is achieved:

$$\forall b \in B,\ \exists a \in A \text{ such that } f(a) = b.$$

Equivalently, $\operatorname{im}(f) = B$.

Remark (Dependencies).

- Function

Definition (Bijective Function)

Definition 3.2.19 (Bijective Function)**.** $f : A \to B$ is *bijective* if it is both injective and surjective.

Remark (Dependencies).

- Function

- Injective Function
- Surjective Function

Remark (Fiber interpretation). Injectivity: each fiber has at most one element. Surjectivity: every fiber is nonempty. Bijectivity: every fiber has exactly one element.

Proposition (One-One Relations and Converse Functionality)

Proposition 3.2.20 (One-One Relations and Converse Functionality)**.** *Let $f : A \to B$ be a function, identified with its graph $G_f \subseteq A \times B$. Then f is injective if and only if the converse relation $G_f^{-1} \subseteq B \times A$ is functional from B to A. Go to proof.*

Remark (Dependencies).

- Injective Function
- Converse Relation
- Functional Relation
- Graph of a Function

Proposition (Bijections Have Unique Correspondence Both Ways)

Proposition 3.2.21 (Bijections Have Unique Correspondence Both Ways)**.** *Let $f : A \to B$ be a function. Then f is bijective if and only if*

$$\forall a \in A\, \exists! b \in B, \quad f(a) = b,$$

and

$$\forall b \in B\, \exists! a \in A, \quad f(a) = b.$$

Go to proof.

Remark (Dependencies).

- Bijective Function
- Injective Function
- Surjective Function

Definition (Many-Place Functional Relation)

Definition 3.2.22 (Many-Place Functional Relation)**.** A relation $R \subseteq A_1 \times \cdots \times A_n \times B$ is *functional in the last coordinate* if

$$\forall a_1 \in A_1 \cdots \forall a_n \in A_n\, \forall b_1, b_2 \in B,$$

$$\big((a_1, \ldots, a_n, b_1) \in R \wedge (a_1, \ldots, a_n, b_2) \in R\big) \Rightarrow b_1 = b_2.$$

When such a relation is also total in the first n coordinates, it determines a function of n variables, written $f(a_1, \ldots, a_n) = b$.

Remark (Dependencies).

- Many-Place Relation
- Functional Relation

Definition (Identity Function)

Definition 3.2.23 (Identity Function)**.** The *identity function* on A is $\mathrm{id}_A : A \to A$, $\mathrm{id}_A(a) = a$.

Remark (Dependencies).

- Function

Definition (Inclusion Map)

Definition 3.2.24 (Inclusion Map)**.** If $A \subseteq B$, the *inclusion map* is $\iota : A \hookrightarrow B$, $\iota(a) = a$.

Remark (Dependencies).

- Function
- Subset

Remark (Identity vs. inclusion). Both send each element to itself, but the inclusion map has a strictly larger codomain when $A \subsetneq B$. The inclusion map is always injective.

Definition (Constant Function)

Definition 3.2.25 (Constant Function)**.** $f : A \to B$ is *constant* if there exists $b_0 \in B$ with $f(a) = b_0$ for all $a \in A$.

Remark (Standard quantified statement). f is constant $\leftrightarrow\ \exists b_0 \in B,\ \forall a \in A,\ f(a) = b_0$.

Remark (Dependencies).

- Function

Composition & Inverses — Quick Reference

Result	Statement	Proof method	Deta
Associativity of $\circ$	$h \circ (g \circ f) = (h \circ g) \circ f$	element-wise	↓ Thr
Identity and $\circ$	$f \circ \mathrm{id}_A = \mathrm{id}_B \circ f = f$	element-wise	↓ Thr
Inj/Surj under $\circ$	Closed under composition; partial converses	element-wise	↓ Thr
Inverse characterization	$f^{-1} \circ f = \mathrm{id}_A$, $f \circ f^{-1} = \mathrm{id}_B$	bijectivity	↓ Thr
Inverse of composition	$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$	verify both sides	↓ Thr
One-sided inverses	Left inverse $\iff$ injective; right $\iff$ surjective		↓ Thr
Preimage preserves ops	f^{-1} commutes with $\cup, \cap, \setminus, c$	element-wise	↓ Thr
Image and set ops	Images preserve $\cup$; contain $\subseteq$ for $\cap$	counterexample	↓ Thr
Image/preimage adjunction	$S \subseteq f^{-1}(f(S))$, $f(f^{-1}(T)) \subseteq T$	element-wise	↓ Thr

3.3 Composition, Inverses, and Set-Image Laws

Definition (Composition)

Definition 3.3.1 (Composition)**.** Let $f : A \to B$ and $g : B \to C$. The *composition* $g \circ f : A \to C$ is:

$$(g \circ f)(a) := g\big(f(a)\big) \quad \text{for all } a \in A.$$

Remark (Dependencies).

- Function
- Cartesian Product

Remark (Non-commutativity). Composition is generally not commutative: $g \circ f \neq f \circ g$ even when both are defined.

Theorem (Associativity of Composition)

Theorem 3.3.2 (Associativity of Composition)**.** *For $f : A \to B$, $g : B \to C$, $h : C \to D$:*

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

Go to proof.

Remark (Standard quantified statement). $\forall f : A \to B,\ g : B \to C,\ h : C \to D,\ \forall a \in A : [h \circ (g \circ f)](a) = [(h \circ g) \circ f](a)$.

Remark (Dependencies).

- Function
- Composition

Theorem (Identity and Composition)

Theorem 3.3.3 (Identity and Composition)**.** *For any $f : A \to B$:*

$$f \circ \mathrm{id}_A = f \quad \textit{and} \quad \mathrm{id}_B \circ f = f.$$

Go to proof.

Remark (Standard quantified statement). $\forall f : A \to B$, $\forall a \in A :\ (f \circ \mathrm{id}_A)(a) = f(a)$ and $(\mathrm{id}_B \circ f)(a) = f(a)$.

Remark (Dependencies).

- Function
- Composition
- Identity Function

Theorem (Injectivity and Surjectivity Under Composition)

Theorem 3.3.4 (Injectivity and Surjectivity Under Composition)**.** *Let $f : A \to B$ and $g : B \to C$.*

(i) If f and g are injective, then $g \circ f$ is injective.

(ii) If f and g are surjective, then $g \circ f$ is surjective.

(iii) If $g \circ f$ is injective, then f is injective.

(iv) If $g \circ f$ is surjective, then g is surjective.

Go to proof.

Remark (Standard quantified statement). (i) $(\forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \to a_1 = a_2) \wedge (\forall b_1, b_2 \in B,\ g(b_1) = g(b_2) \to b_1 = b_2)\ \to\ \forall a_1, a_2 \in A,\ g(f(a_1)) = g(f(a_2)) \to a_1 = a_2$. (iii) $(\forall a_1, a_2 \in A,\ g(f(a_1)) = g(f(a_2)) \to a_1 = a_2)\ \to\ \forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \to a_1 = a_2$.

Remark (Dependencies).

- Function
- Composition
- Injective Function
- Surjective Function

Remark (Sharpness of the implications). In (iii), g need not be injective. In (iv), f need not be surjective. Bijectivity of $g \circ f$ does not imply bijectivity of either f or g individually.

Definition (Inverse Function)

Definition 3.3.5 (Inverse Function)**.** Let $f : A \to B$ be bijective. The *inverse function* is $f^{-1} : B \to A$ defined by: $f^{-1}(b) = a$ iff $f(a) = b$.

Remark (Dependencies).

- Function
- Bijective Function
- Identity Function

Theorem (Characterization of Inverse Functions)

Theorem 3.3.6 (Characterization of Inverse Functions)**.** *Let $f : A \to B$ be bijective. Then*

$$f^{-1} \circ f = \mathrm{id}_A \quad \text{and} \quad f \circ f^{-1} = \mathrm{id}_B\,.$$

Conversely, a function admits an inverse iff it is bijective. *Go to proof.*

Remark (Standard quantified statement). $\forall a \in A : \quad f^{-1}(f(a)) = a$ and $\forall b \in B : f(f^{-1}(b)) = b$.
Converse: $(\exists g : B \to A,\ g \circ f = \mathrm{id}_A \wedge f \circ g = \mathrm{id}_B) \ \leftrightarrow \ f$ is bijective.

Remark (Dependencies).

- Function
- Composition
- Inverse Function
- Bijective Function
- Identity Function

Theorem (Inverse of a Composition)

Theorem 3.3.7 (Inverse of a Composition)**.** *Let $f : A \to B$ and $g : B \to C$ be bijective. Then $g \circ f$ is bijective and*

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Go to proof.

Remark (Standard quantified statement). $\forall c \in C : \ (g \circ f)^{-1}(c) = f^{-1}(g^{-1}(c))$.

Remark (Dependencies).

- Function
- Composition
- Inverse Function
- Bijective Function

Definition (Left Inverse)

Definition 3.3.8 (Left Inverse)**.** Let $f : A \to B$. A *left inverse* of f is a function $g : B \to A$ with $g \circ f = \mathrm{id}_A$.

Remark (Dependencies).

- Function
- Composition
- Identity Function

Definition (Right Inverse)

Definition 3.3.9 (Right Inverse)**.** Let $f : A \to B$. A *right inverse* (or *section*) of f is a function $h : B \to A$ with $f \circ h = \mathrm{id}_B$.

Remark (Dependencies).

- Function
- Composition
- Identity Function

Remark (One-sided inverses). Left inverses and right inverses are the two one-sided inverse notions.

Theorem (One-Sided Inverses and Function Properties)

Theorem 3.3.10 (One-Sided Inverses and Function Properties)**.** *Let $f : A \to B$.*

(i) f has a left inverse $\iff$ f is injective (and $A \neq \varnothing$ or $A = B = \varnothing$).

(ii) f has a right inverse $\iff$ f is surjective.

(iii) If f has both a left inverse g and a right inverse h, then $g = h$ and f is bijective.

Go to proof.

Remark (Standard quantified statement). (i) $(\exists g : B \to A,\ \forall a \in A,\ g(f(a)) = a) \leftrightarrow \forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \to a_1 = a_2$.
(ii) $(\exists h : B \to A,\ \forall b \in B,\ f(h(b)) = b) \leftrightarrow \forall b \in B,\ \exists a \in A,\ f(a) = b$.

Remark (Dependencies).

- Function
- Composition
- Left Inverse
- Right Inverse
- Injective Function
- Surjective Function
- Bijective Function

Remark (Axiom of Choice). The existence of a right inverse for every surjective function is equivalent to the Axiom of Choice.

Definition (Restriction)

Definition 3.3.11 (Restriction)**.** Let $f : A \to B$ and $C \subseteq A$. The *restriction* of f to C is $f|_C : C \to B$, $f|_C(a) = f(a)$ for all $a \in C$.

Remark (Dependencies).

- Function
- Subset
- Domain

Definition (Extension)

Definition 3.3.12 (Extension)**.** Let $A \subseteq A'$, and let $f : A \to B$. A function $g : A' \to B$ is an *extension* of f if $g|_A = f$.

Remark (Dependencies).

- Function
- Subset
- Restriction
- Domain
- Codomain

Remark (Extensions not unique). An extension agrees with f on all of A but may be defined on a larger domain A'. Extensions are generally not unique; uniqueness requires additional constraints (such as continuity in analysis or linearity in algebra), leading to important results like the Tietze Extension Theorem and the Hahn–Banach Theorem.

Theorem (Preimages Preserve Set Operations)

Theorem 3.3.13 (Preimages Preserve Set Operations)**.** *Let $f : A \to B$ and $S, T \subseteq B$. Then*

(i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$,

(ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$,

(iii) $f^{-1}(S \setminus T) = f^{-1}(S) \setminus f^{-1}(T)$,

(iv) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Go to proof.

Remark (Standard quantified statement). (i) $\forall a \in A : \ a \in f^{-1}(S \cup T) \leftrightarrow (f(a) \in S \vee f(a) \in T)$.
(ii) $\forall a \in A : \ a \in f^{-1}(S \cap T) \leftrightarrow (f(a) \in S \wedge f(a) \in T)$.
(iv) $\forall a \in A : \ a \in f^{-1}(S^c) \leftrightarrow f(a) \notin S$.

Remark (Dependencies).

- Function
- Preimage
- Union
- Intersection
- Set Difference
- Complement

Theorem (Images and Set Operations)

Theorem 3.3.14 (Images and Set Operations)**.** *Let $f : A \to B$ and $S, T \subseteq A$. Then*

(i) $f(S \cup T) = f(S) \cup f(T)$,

(ii) $f(S \cap T) \subseteq f(S) \cap f(T)$,

(iii) $f(S \setminus T) \supseteq f(S) \setminus f(T)$.

Equality holds in (ii) and (iii) for all S, T iff f is injective. *Go to proof.*

Remark (Standard quantified statement)*.* (i) $\forall b \in B : \; b \in f(S \cup T) \leftrightarrow (\exists a \in S, \; f(a) = b) \vee (\exists a \in T, \; f(a) = b)$.
(ii) $\forall b \in B : \; (\exists a \in S \cap T, \; f(a) = b) \to (\exists a \in S, \; f(a) = b) \wedge (\exists a \in T, \; f(a) = b)$; equality iff f injective.

Remark (Dependencies)*.*

- Function
- Image of a Set
- Union
- Intersection
- Set Difference
- Injective Function

Theorem (Image–Preimage Adjunction)

Theorem 3.3.15 (Image–Preimage Adjunction)**.** *Let $f : A \to B$, let $S \subseteq A$, and let $T \subseteq B$. Then*

$$S \subseteq f^{-1}(f(S)), \qquad f(f^{-1}(T)) \subseteq T.$$

Moreover, $S = f^{-1}(f(S))$ for every $S \subseteq A$ iff f is injective, and $f(f^{-1}(T)) = T$ for every $T \subseteq B$ iff f is surjective. *Go to proof.*

Remark (Dependencies)*.*

- Function
- Image of a Set
- Preimage
- Injective Function
- Surjective Function

Remark (Images, preimages, and Boolean operations).

Operation	**Preimage law, for $S, T \subseteq B$**	**Image law, for $C, D \subseteq A$**
Union	$f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$	$f(C \cup D) = f(C) \cup f(D)$
Intersection	$f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$	$f(C \cap D) \subseteq f(C) \cap f(D)$; equality for all C, D iff f is injective
Difference	$f^{-1}(S \setminus T) = f^{-1}(S) \setminus f^{-1}(T)$	$f(C \setminus D) \supseteq f(C) \setminus f(D)$; equality for all C, D iff f is injective
Complement	$f^{-1}(B \setminus S) = A \setminus f^{-1}(S)$	In general, $f(A \setminus C)$ need not equal $B \setminus f(C)$

Remark (Composition and inverse interaction).

Construction	**Set-operation interaction**	**Reference**
Composition	If $f : A \to B$, $g : B \to C$, and $U \subseteq C$, then $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$.	Composition, Preimage
Inverse function	If $f : A \to B$ is bijective, inverse images along f^{-1} are ordinary preimages for the function $f^{-1} : B \to A$.	Inverse Function, Inverse Characterization
Inverse of composition	If f and g are bijective, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$, so inverse order reverses composition.	Inverse of a Composition
One-sided inverse	A left inverse detects injectivity; a right inverse detects surjectivity. These properties control when image inclusions become equalities.	One-Sided Inverses

Remark (Asymmetry between images and preimages). Preimages preserve union, intersection, difference, and complement exactly. Forward images preserve unions exactly, but generally do not preserve intersections, differences, or complements. The inclusions for intersections and differences become equalities uniformly when the function is injective. This asymmetry is fundamental and explains why preimages appear more naturally in topology and measure theory.

Chapter 4

Orderings

orderings

Functions → **Orderings** → Cardinality

4.1 Order

Ordered Sets — Quick Reference

Concept	Meaning	Detail
Ordered set	$(A, \leq)$: set with a partial order	↓ Def
Strict order	$a < b \iff a \leq b \wedge a \neq b$	↓ Def
Comparable / incomparable	$a \leq b$ or $b \leq a$ / neither	↓ Def
Total (linear) order	Partial order with all pairs comparable	↓ Def
Upper / lower bound	$u \geq s$ / $\ell \leq s$ for all $s \in S$	↓ Def
Minimal / maximal	No smaller/larger element in S	↓ Def
Least / greatest	Smaller/larger than all elements of S	↓ Def
Order-preserving	$a \leq b \Rightarrow f(a) \leq' f(b)$	↓ Def
Order isomorphism	Bijective order-preserving map with order-preserving inverse	↓ Def
Well-ordered set	Every nonempty subset has a least element	↓ Def
Chain / antichain	All comparable / none comparable	↓ Def
Initial segment	Downward-closed subset	↓ Def

4.2 Ordered Sets

Definition (Ordered Set)

Definition 4.2.1 (Ordered Set)**.** An *ordered set* (or *partially ordered set*, *poset*) is a pair $(A, \leq)$ where $\leq$ is a partial order on A: a relation that is reflexive, antisymmetric, and transitive.

Remark (Dependencies).

- Partial Order

Remark (English reading). An ordered set is a set equipped with a specified way to compare elements, but without requiring all pairs to be comparable.

Definition (Strict Order)

Definition 4.2.2 (Strict Order)**.** Let $(A, \leq)$ be an ordered set. The *strict order* $<$ is defined by:

$$a < b \iff (a \leq b \wedge a \neq b).$$

Remark (Dependencies).

- Ordered Set
- Partial Order

Remark (Strict and non-strict are equivalent). The relation $<$ is irreflexive and transitive. Conversely, given any strict partial order $<$, one recovers a non-strict order by setting $a \leq b \iff (a < b \vee a = b)$. The two formulations carry exactly the same information.

Definition (Comparable and Incomparable Elements)

Definition 4.2.3 (Comparable and Incomparable Elements)**.** In an ordered set $(A, \leq)$, elements $a, b \in A$ are *comparable* if $a \leq b$ or $b \leq a$; they are *incomparable* if neither holds.

Remark (Standard quantified statement). a, b comparable $\leftrightarrow$ $a \leq b \vee b \leq a$. a, b incomparable $\leftrightarrow$ $\neg(a \leq b) \wedge \neg(b \leq a)$.

Remark (Incomparable elements). Incomparable elements can only occur in partial orders. In a total order, every pair is comparable by definition.

Definition (Total / Linear Order)

Definition 4.2.4 (Total / Linear Order)**.** An ordered set $(A, \leq)$ is a *total order* (or *linear order*) if every pair of elements is comparable:

$$\forall a, b \in A,\ a \leq b \vee b \leq a.$$

Remark (Examples). $(\mathbb{R}, \leq)$ is a total order. $(\mathcal{P}(A), \subseteq)$ is a partial order that is generally not total.

Definition (Upper and Lower Bounds)

Remark (Dependencies).

- Ordered Set
- Comparable and Incomparable Elements
- Total Order

Definition 4.2.5 (Upper and Lower Bounds)**.** Let $(A, \leq)$ be an ordered set and $S \subseteq A$.
An element $u \in A$ is an *upper bound* of S if $\forall s \in S,\ s \leq u$.
An element $\ell \in A$ is a *lower bound* of S if $\forall s \in S,\ \ell \leq s$.

Remark (Dependencies).

- Ordered Set
- Subset

Remark (Consequence). Bounds need not exist and need not be unique. They need not lie in S itself. Bounds are central to the definition of completeness for ordered fields.

Definition (Minimal, Maximal, Least, Greatest Elements)

Let $S \subseteq A$ in an ordered set $(A, \leq)$.
$m \in S$ is a *minimal element* of S if $\nexists s \in S$ with $s < m$.
$M \in S$ is a *maximal element* of S if $\nexists s \in S$ with $M < s$.

$\ell \in S$ is the *least element* if $\forall s \in S,\ \ell \leq s$.
$g \in S$ is the *greatest element* if $\forall s \in S,\ s \leq g$.

Remark (Minimal vs. least). A *least* element is below every element of S; it must be unique if it exists. A *minimal* element merely has nothing below it within S; there may be many. Every least element is minimal, but not conversely.

Definition (Order-Preserving Map and Isomorphism)

Let $(M, \leq)$ and $(M', \leq')$ be partially ordered sets.
A function $f : M \to M'$ is *order-preserving* (or *monotone*) if

$$\forall a, b \in M, \quad a \leq b \implies f(a) \leq' f(b).$$

f is an *order isomorphism* if it is bijective and

$$\forall a, b \in M, \quad a \leq b \iff f(a) \leq' f(b).$$

Remark (Order-isomorphic posets). An order isomorphism preserves and reflects the order structure exactly, including comparability, minimal and maximal elements, and bounds. Two posets are *order-isomorphic* if such a map exists; they are structurally identical from the order-theoretic viewpoint.

Definition (Well-Ordered Set)

Definition 4.2.6 (Well-Ordered Set). An ordered set $(A, <)$ is *well-ordered* if every nonempty subset $S \subseteq A$ has a least element:

$$\forall S \subseteq A,\ \big(S \neq \varnothing \ \Rightarrow\ \exists m \in S \text{ s.t. } m \leq s\ \forall s \in S\big).$$

Remark (Dependencies).

- Ordered Set
- Subset
- Total Order

Remark (Well-order implies total order). Every well-ordered set is totally ordered. The well-ordering condition is strictly stronger: it requires a least element in every nonempty subset, not just in A as a whole.

Example (Well-order examples). $(\mathbb{N}, \leq)$ is well-ordered. $(\mathbb{Z}, \leq)$ is not: the set $\{\ldots, -3, -2, -1\}$ has no least element. $(\mathbb{R}, \leq)$ is not: $(0, 1)$ has no least element.

Remark (Connection to ordinal numbers). An *ordinal number* is defined as an equivalence class of well-ordered sets under order isomorphism: it measures the *order type* of a well-ordered set, not merely its cardinality.

Definition (Chain and Antichain)

Definition 4.2.7 (Chain and Antichain)**.** A subset $C \subseteq A$ of a poset $(A, \leq)$ is a *chain* if every pair of elements in C is comparable. It is an *antichain* if no two distinct elements of C are comparable.

Remark (Standard quantified statement). C is a chain $\leftrightarrow$ $\forall a, b \in C,\ a \leq b \vee b \leq a$. C is an antichain $\leftrightarrow$ $\forall a, b \in C,\ a \neq b \rightarrow \neg(a \leq b) \wedge \neg(b \leq a)$.

Definition (Initial Segment)

Definition 4.2.8 (Initial Segment)**.** A subset $I \subseteq A$ is an *initial segment* of $(A, \leq)$ if

$$a \in I \text{ and } b \leq a \ \Rightarrow\ b \in I.$$

Remark (Standard quantified statement). I is an initial segment $\leftrightarrow$ $\forall a \in I,\ \forall b \in A,\ b \leq a \rightarrow b \in I$.

Remark (Transition). Ordered sets provide the abstract framework for order structures on the real numbers, function spaces, and metric spaces. In later sections, order interacts with topology and analysis through intervals, monotone functions, and the completeness axiom for $\mathbb{R}$.

Definition (Directed Set)

Definition 4.2.9 (Directed Set)**.** A preorder $(I, \leq)$ is *directed* (or a *directed set*) if every finite subset of I has an upper bound in I: for all $i, j \in I$ there exists $k \in I$ with $i \leq k$ and $j \leq k$.

Remark (Dependencies).

- Partial Order
- Upper and Lower Bounds
- Finite and Infinite Sets

Remark (What directed means). In a directed set, any two elements can always be "dominated" by a third. The condition does not require a maximum element to exist, only that no two elements are ever incomparable in a final way: there is always a common upper bound available. Every totally ordered set is directed (take $k = \max(i, j)$), but directed sets can be far more general.

Example (Canonical directed sets).

(i) $(\mathbb{N}, \leq)$ is directed: given $m, n \in \mathbb{N}$, take $k = \max(m, n)$.

(ii) The collection of all finite subsets of a set X, ordered by inclusion, is directed: given finite sets $F_1, F_2 \subseteq X$, take $k = F_1 \cup F_2$.

(iii) The set of all open neighborhoods of a point x in a topological space, ordered by reverse inclusion ($U \leq V$ iff $U \supseteq V$), is directed: given neighborhoods U and V of x, take $k = U \cap V$, which is also a neighborhood of x.

(iv) A partially ordered set is directed if and only if every finite subset has an upper bound. Partially ordered sets that are *not* directed have "incomparable maximal elements" — elements with no common upper bound.

Definition (Net)

Definition 4.2.10 (Net)**.** Let $(I, \leq)$ be a directed set and X a set. A *net* in X is a function $x : I \to X$. The value at $i \in I$ is written x_i, and the net is written $(x_i)_{i \in I}$.

Remark (Dependencies).

- Directed Set
- Function

Remark (Nets generalize sequences). A sequence is a net with index set $(\mathbb{N}, \leq)$. Nets are the correct generalization of sequences to general topological spaces: in a metric space, sequences suffice to detect all topological properties (closure, continuity, compactness). In a general topological space, sequences can fail to detect these properties — one needs nets. Specifically, a subset A of a topological space is closed if and only if every convergent net in A has its limit in A; and a function is continuous if and only if it preserves limits of nets.

The definition of net is placed here because it is purely order-theoretic: a directed index set and a function. The analytic content — what it means for a net to converge — belongs to topology and functional analysis. By establishing the set-theoretic definition now, those chapters can immediately state convergence conditions without set-theoretic digression.

Supremum, Infimum, and Bounds — Quick Reference

Concept	Meaning	Detail
Upper bound	$u \geq s$ for all $s \in S$	↓ Def
Lower bound	$\ell \leq s$ for all $s \in S$	↓ Def
Supremum	Least upper bound of S	↓ Def
Infimum	Greatest lower bound of S	↓ Def
Uniqueness	Suprema and infima are unique when they exist	↓ Prop
Max/min vs. sup/inf	Maximum $\Rightarrow$ supremum; not conversely	↓ Rem
Duality	sup in $(A, \leq)$ = inf in $(A, \geq)$	↓ Prop

4.3 Supremum and Infimum

Every student of analysis encounters the least upper bound property of $\mathbb{R}$ early and prominently. But this property is not a peculiarity of real numbers: it is an order-theoretic concept that can be formulated in any partially ordered set. Before we reach $\mathbb{R}$, it is essential to understand supremum and infimum in their natural habitat — abstract ordered sets — so that when the completeness axiom appears in Volume II, you will recognise exactly what it is asserting and why $\mathbb{Q}$ fails to satisfy it.

The definitions in this section are also the backbone of measure theory. The extended real line $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$, equipped with the usual order, is the ambient poset in which measures take their values. Countable additivity, the monotone convergence theorem, and Fatou's lemma are all statements about suprema of sequences. Understanding supremum and infimum at the abstract level here means those later theorems will feel inevitable rather than arbitrary.

The definition of upper and lower bound was given in the Ordered Sets section. For convenience we recall: $u \in A$ is an *upper bound* of $S \subseteq A$ if $\forall s \in S,\ s \leq u$; a *lower bound* satisfies the reverse. We say S is *bounded above* (resp. *bounded below*, *bounded*) if it has at least one upper (resp. lower, both) bound.

Remark (Bounds depend on the ambient poset)*.* Whether S is bounded above depends on the poset $(A, \leq)$, not on S alone. The set $\{r \in \mathbb{Q} : r^2 < 2\}$ is bounded above in $(\mathbb{Q}, \leq)$ — the rational number 2 is an upper bound — but it has *no* supremum inside $\mathbb{Q}$. The same set in $(\mathbb{R}, \leq)$ has supremum $\sqrt{2} \in \mathbb{R}$. Being bounded above is necessary but not sufficient for a supremum to exist; the completeness axiom for $\mathbb{R}$ is precisely the assertion that sufficiency holds in $\mathbb{R}$.

Definition (Supremum and Infimum)

Definition 4.3.1 (Supremum and Infimum)**.** Let $(A, \leq)$ be a poset and $S \subseteq A$. The *supremum* (or *least upper bound*) of S, written $\sup S$, is an element $u^* \in A$ satisfying:

(i) u^* is an upper bound of S: $\forall s \in S,\ s \leq u^*$;

(ii) u^* is the *least* upper bound: if u is *any* upper bound of S, then $u^* \leq u$.

The *infimum* (or *greatest lower bound*) of S, written $\inf S$, is an element $\ell^* \in A$ satisfying:

(i) ℓ^* is a lower bound of S: $\forall s \in S,\ \ell^* \leq s$;

(ii) ℓ^* is the *greatest* lower bound: if ℓ is *any* lower bound of S, then $\ell \leq \ell^*$.

Remark (Standard quantified statement)*.* $u^* = \sup S \ \leftrightarrow\ (\forall s \in S,\ s \leq u^*) \ \wedge\ (\forall u \in A,\ (\forall s \in S,\ s \leq u) \ \to\ u^* \leq u)$.
$\ell^* = \inf S \ \leftrightarrow\ (\forall s \in S,\ \ell^* \leq s) \ \wedge\ (\forall \ell \in A,\ (\forall s \in S,\ \ell \leq s) \ \to\ \ell \leq \ell^*)$.

Remark (Dependencies)*.*

- Upper and Lower Bounds
- Partial Order
- Subset

Remark (How to read these definitions)*.* Condition (i) says u^* is a *candidate*: it stands above everything in S. Condition (ii) says u^* is the *best* candidate: it is at most as large as any other upper bound. Together they say: u^* is the unique element of A that is simultaneously an upper bound of S and no larger than any other upper bound.

A useful restatement of condition (ii) is its contrapositive: if $v < u^*$, then v is *not* an upper bound of S, meaning there exists some $s \in S$ with $v < s$. This is how the condition is typically used in proofs in analysis: to verify that $u^* = \sup S$, you show (i) u^* is an upper bound, and (ii) for every $\varepsilon > 0$, the value $u^* - \varepsilon$ is *not* an upper bound, i.e., there exists $s \in S$ with $s > u^* - \varepsilon$.

Proposition

Proposition 4.3.2 (Uniqueness of Supremum and Infimum). *Let $(A, \leq)$ be a poset and $S \subseteq A$. If $\sup S$ exists, it is unique. If $\inf S$ exists, it is unique.*

Remark (Standard quantified statement). $\forall u^*, v^* \in A$: if both u^* and v^* satisfy the two conditions in the definition of $\sup S$, then $u^* = v^*$.

Remark (Dependencies).

- Supremum and Infimum
- Partial Order
- Subset

Remark (Proof idea). If u^* and v^* are both suprema, then u^* is an upper bound, so the minimality of v^* gives $v^* \leq u^*$; and symmetrically $u^* \leq v^*$. Antisymmetry of the partial order then forces $u^* = v^*$. This is a standard antisymmetry argument: when two elements are mutually $\leq$-related in a partial order, they must be equal.

Remark (Supremum need not belong to S). The set $S = (0, 1) \subseteq \mathbb{R}$ is open: it contains neither endpoint. Yet $\sup S = 1$ and $\inf S = 0$. Neither 0 nor 1 belongs to S.

When the supremum does happen to lie in S, it is called the *maximum* of S, written $\max S$. Thus $\max S = \sup S$ when $\sup S \in S$, and $\max S$ is undefined when $\sup S \notin S$. Every maximum is a supremum, but the converse fails: a supremum may exist without being a maximum. The same relationship holds between infimum and minimum.

This distinction is critical in analysis. A continuous function on a closed bounded interval always achieves its maximum (the Extreme Value Theorem), but on an open interval it may approach a supremum without ever attaining it.

Remark (Supremum need not exist). Being bounded above is necessary but not sufficient for a supremum to exist. Consider $S = \{r \in \mathbb{Q} : r^2 < 2\}$ in the poset $(\mathbb{Q}, \leq)$. Every element of S satisfies $r < 2$, so $2 \in \mathbb{Q}$ is an upper bound. But the candidate supremum would need to be $\sqrt{2}$, and $\sqrt{2} \notin \mathbb{Q}$. Any rational upper bound $q > \sqrt{2}$ is not the least upper bound, because $\frac{q+\sqrt{2}}{2}$ is a smaller upper bound (and is also irrational, which means we can always find a rational strictly between any rational upper bound and $\sqrt{2}$, and that rational is still an upper bound). In short: S is bounded above in $\mathbb{Q}$, but $\sup S$ does not exist in $\mathbb{Q}$.

This is the precise failure that the *completeness axiom* for $\mathbb{R}$ corrects: it asserts that every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. The rational numbers satisfy every other ordered field axiom but fail this one.

Example (Sup and inf in divisibility). Order the set $D = \{1, 2, 3, 4, 6, 12\}$ by divisibility: $a \le b$ iff $a \mid b$. Take $S = \{4, 6\}$.

Upper bounds of S: an element $u \in D$ must satisfy $4 \mid u$ and $6 \mid u$, i.e., u must be a common multiple of 4 and 6 in D. The only such element is 12, so $\sup S = 12 = \operatorname{lcm}(4, 6)$.

Lower bounds of S: an element $\ell \in D$ must satisfy $\ell \mid 4$ and $\ell \mid 6$, i.e., ℓ must be a common divisor of 4 and 6 in D. The common divisors are $\{1, 2\}$; the greatest is 2, so $\inf S = 2 = \gcd(4, 6)$.

This example reveals a deep pattern: in any divisibility poset, sup computes the least common multiple and inf computes the greatest common divisor. Order theory and number theory are speaking the same language.

Example (Sup and inf for function spaces). Let $B(X)$ be the set of bounded real-valued functions on a set X, ordered pointwise: $f \le g$ iff $f(x) \le g(x)$ for all $x \in X$. For a collection $\{f_\alpha\}$ of bounded functions, the pointwise supremum is $(\sup_\alpha f_\alpha)(x) = \sup_\alpha f_\alpha(x)$. In measure theory and integration theory, one constantly takes suprema and infima of sequences of functions; this example shows that such operations are instances of the abstract sup/inf defined here.

Proposition

Proposition 4.3.3 (Characterisation of Supremum). *Let $(A, \le)$ be a poset, $S \subseteq A$, and $u^* \in A$. Then $u^* = \sup S$ if and only if:*

(i) u^* is an upper bound of S; and

(ii) for every $v \in A$ with $v < u^*$, there exists $s \in S$ with $v < s$.

Remark (Standard quantified statement). $u^* = \sup S \;\leftrightarrow\; (\forall s \in S,\ s \le u^*) \;\wedge\; (\forall v \in A,\ v < u^* \;\rightarrow\; \exists s \in S,\ v < s)$.

Remark (Dependencies).

- Supremum and Infimum
- Partial Order
- Upper and Lower Bounds
- Subset

Remark (Why this characterisation is useful). Condition (ii) here is the contrapositive of the minimality condition in the definition. In analysis this is the working form: to check that u^* is the supremum, you verify that nothing strictly below u^* is an upper bound. In $\mathbb{R}$ this becomes: for every $\varepsilon > 0$, there exists $s \in S$ with $s > u^* - \varepsilon$. This is the exact form used in the proof that sup interacts correctly with limits, integrals, and measures.

Proposition

Proposition 4.3.4 (Duality of Supremum and Infimum). *Let $(A, \le)$ be a poset and $S \subseteq A$. Then $\inf_{(A,\le)} S = \sup_{(A,\ge)} S$, where $(A, \ge)$ denotes the dual poset.*
In particular: if every nonempty bounded-above subset of $(A, \le)$ has a supremum, then every nonempty bounded-below subset of $(A, \le)$ has an infimum.

Remark (Standard quantified statement). $\forall S \subseteq A$: ℓ^* is the infimum of S in $(A, \leq)$ $\leftrightarrow$ ℓ^* is the supremum of S in $(A, \geq)$.

Remark (Dependencies).

- Supremum and Infimum
- Partial Order
- Duality Principle for Posets
- Subset

Remark (Consequence for $\mathbb{R}$). This proposition explains why the completeness axiom for $\mathbb{R}$ is stated only for suprema: the infimum statement comes for free by duality. The axiom says every nonempty bounded-above subset has a supremum; the proposition immediately gives that every nonempty bounded-below subset has an infimum. You will use this constantly without comment in Volume II.

Remark (Sequences and sup/inf). A sequence $(a_n)_{n=1}^{\infty}$ in $\mathbb{R}$ is a function $\mathbb{N} \to \mathbb{R}$, so its range $\{a_n : n \in \mathbb{N}\}$ is a subset of $\mathbb{R}$. The quantities $\sup_n a_n$ and $\inf_n a_n$ are the supremum and infimum of this range set in the poset $(\mathbb{R}, \leq)$.

More subtle are the *limit superior* and *limit inferior*:

$$\limsup_{n\to\infty} a_n = \inf_{n\geq 1} \sup_{k\geq n} a_k, \qquad \liminf_{n\to\infty} a_n = \sup_{n\geq 1} \inf_{k\geq n} a_k.$$

These are iterated suprema and infima. The fact that they always exist in $\overline{\mathbb{R}}$ (the extended reals, where $\pm\infty$ are adjoined to supply missing bounds) is a direct application of the completeness of $\mathbb{R}$. Mastering the abstract definitions here makes those formulas transparent rather than mysterious.

Remark (Transition). Supremum and infimum are defined for arbitrary subsets of a poset, but they need not always exist. When *every* two-element subset $\{a, b\}$ has both a supremum and an infimum, the poset has enough structure to define binary operations $a \vee b = \sup\{a, b\}$ and $a \wedge b = \inf\{a, b\}$. This additional structure is called a *lattice*, and it is developed in the next section. Lattices are the precise abstract setting for the σ-algebras and topologies encountered in measure theory and topology.

Lattices — Quick Reference

Concept	**Meaning**	**Detail**
Lattice	Poset with $\sup\{a, b\}$ and $\inf\{a, b\}$ for all a, b	↓ Def
Join / Meet	$a \vee b = \sup\{a, b\}$; $a \wedge b = \inf\{a, b\}$	↓ Def
Complete lattice	Every subset has sup and inf	↓ Def
Distributive lattice	$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$	↓ Def
Boolean lattice	Complemented distributive lattice	↓ Def
$\mathcal{P}(X)$	Complete Boolean lattice under $\subseteq$	↓ Ex
σ-algebras	Sub-lattices of $\mathcal{P}(X)$ closed under countable ops	↓ Rem
Knaster–Tarski	Every monotone map on a complete lattice has a fixed point	↓ Thm

4.4 Lattices

The preceding section defined supremum and infimum for arbitrary subsets of a poset. In many posets that appear in practice, every pair of elements already has a supremum and an infimum. When this is the case, the binary operations $a \vee b = \sup\{a, b\}$ and $a \wedge b = \inf\{a, b\}$ are well-defined everywhere, giving the poset the algebraic structure of a *lattice*.

Lattices are ubiquitous:

- The power set $\mathcal{P}(X)$ ordered by $\subseteq$ is a complete lattice, with join = union and meet = intersection.
- Every σ-algebra is a sub-lattice of $\mathcal{P}(X)$ that is additionally closed under countable joins and meets and under complementation.
- Every topology is a sub-structure of $\mathcal{P}(X)$ that is closed under arbitrary joins and finite meets.
- In functional analysis, the lattice of closed subspaces of a Hilbert space is a complete lattice under inclusion.

Understanding lattices here means that when σ-algebras, topologies, and sub-σ-algebras are encountered downstream, their closure conditions will be recognisable instances of a single abstract pattern.

Definition (Lattice)

Definition 4.4.1 (Lattice)**.** A *lattice* is a poset $(L, \leq)$ in which every two-element subset $\{a, b\} \subseteq L$ has both a supremum and an infimum in L.
The supremum $\sup\{a, b\}$ is called the *join* of a and b, written $a \vee b$. The infimum $\inf\{a, b\}$ is called the *meet* of a and b, written $a \wedge b$.

Remark (Standard quantified statement)*.* $(L, \leq)$ is a lattice $\leftrightarrow$ $\forall a, b \in L$, $\exists\, a \vee b \in L$ and $\exists\, a \wedge b \in L$, where $a \vee b$ and $a \wedge b$ satisfy the sup and inf conditions of Definition (Supremum and Infimum).

Remark (Join and meet as operations)*.* In a lattice, $\vee$ and $\wedge$ are binary operations $L \times L \to L$. They satisfy, for all $a, b, c \in L$:

(i) *Commutativity*: $a \vee b = b \vee a$ and $a \wedge b = b \wedge a$.

(ii) *Associativity*: $(a \vee b) \vee c = a \vee (b \vee c)$ and $(a \wedge b) \wedge c = a \wedge (b \wedge c)$.

(iii) *Idempotency*: $a \vee a = a$ and $a \wedge a = a$.

(iv) *Absorption*: $a \vee (a \wedge b) = a$ and $a \wedge (a \vee b) = a$.

Properties (i)–(iv) follow directly from the order-theoretic definitions of join and meet. Conversely, any set with two binary operations satisfying (i)–(iv) can be given a partial order making it a lattice: define $a \leq b$ iff $a \wedge b = a$. The two approaches — order-theoretic and algebraic — are equivalent.

Example (Power set lattice). For any set X, the power set $(\mathcal{P}(X), \subseteq)$ is a lattice: $A \vee B = A \cup B$ and $A \wedge B = A \cap B$. The algebraic laws of the Set Algebra section (commutativity, associativity, distributivity, absorption) are exactly the lattice laws in this example. This is not a coincidence: Boolean algebra is a special kind of lattice.

Example (Divisibility lattice). The divisibility order on $\mathbb{N}^+ = \{1, 2, 3, \ldots\}$ makes it a lattice: $a \vee b = \operatorname{lcm}(a, b)$ and $a \wedge b = \gcd(a, b)$, as we noted in Example 4.3.

Example (Not every poset is a lattice). Consider the poset $(\{a, b, c, d\}, \leq)$ with $a < c$, $a < d$, $b < c$, $b < d$, and a, b incomparable. The subset $\{a, b\}$ has two minimal upper bounds, c and d, which are incomparable. Hence $\sup\{a, b\}$ does not exist, and this poset is not a lattice.

The power set $\mathcal{P}(\{1, 2, 3\})$ restricted to subsets of cardinality $\neq 1$ provides a cleaner example: $\{1, 2\}$ and $\{1, 3\}$ have no meet in this restricted collection.

Definition (Complete Lattice)

Remark (Dependencies).

- Partial Order
- Subset
- Supremum and Infimum

Definition 4.4.2 (Complete Lattice)**.** A lattice $(L, \leq)$ is a *complete lattice* if every subset $S \subseteq L$ (including the empty set and L itself) has both a supremum and an infimum in L:

$$\forall S \subseteq L, \quad \sup S \in L \text{ and } \inf S \in L.$$

Remark (Standard quantified statement). $(L, \leq)$ is a complete lattice $\leftrightarrow$ $\forall S \subseteq L$, $\exists u^* \in L$, $\exists \ell^* \in L$ satisfying the sup and inf conditions of Definition (Supremum and Infimum).

Remark (The empty set forces top and bottom). When $S = \varnothing$, the sup condition says: u^* is an upper bound of the empty set (which is vacuously satisfied by every element of L), and u^* is at most every other upper bound. This forces $u^* = \inf L$, the *bottom element* of L (often written $\perp$). Symmetrically, $\inf \varnothing = \sup L$, the *top element* (often written $\top$).

Concretely: in $(\mathcal{P}(X), \subseteq)$, we have $\sup \varnothing = \varnothing$ (the empty union) and $\inf \varnothing = X$ (the empty intersection equals the whole universe, by convention). Every complete lattice therefore has a top element $\top = \sup L$ and a bottom element $\perp = \inf L$.

Example ($\mathcal{P}(X)$ is a complete lattice). For any set X, the power set $(\mathcal{P}(X), \subseteq)$ is a complete lattice: for any family $\{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$,

$$\sup_{\alpha \in I} A_\alpha = \bigcup_{\alpha \in I} A_\alpha, \qquad \inf_{\alpha \in I} A_\alpha = \bigcap_{\alpha \in I} A_\alpha.$$

The bottom element is $\varnothing$ and the top element is X. This is the canonical example of a complete lattice, and every σ-algebra on X is a complete sub-lattice of $(\mathcal{P}(X), \subseteq)$.

Example ($[0, \infty]$ is a complete lattice). The extended positive half-line $[0, \infty] = [0, \infty) \cup \{+\infty\}$ with the usual order is a complete lattice. Every nonempty subset has a supremum (which may be $+\infty$) and an infimum (which may be 0). This is the value space for measures: a measure assigns to each measurable set a value in $[0, \infty]$, and the key properties of measures (countable additivity, monotone convergence) are stated in terms of sups and infs in this complete lattice.

Definition (Distributive Lattice)

Remark (Dependencies).

- Lattice
- Subset
- Supremum and Infimum

Definition 4.4.3 (Distributive Lattice)**.** A lattice $(L, \leq)$ is *distributive* if for all $a, b, c \in L$:

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c).$$

Remark (Dependencies).

- Lattice

Remark (One law implies the other)*.* By the duality principle for lattices, distributivity of $\wedge$ over $\vee$ is equivalent to distributivity of $\vee$ over $\wedge$: $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$. Proving one gives the other for free.

Remark (Why distributivity matters)*.* The distributive law is what makes a lattice behave like set algebra. In the power set $\mathcal{P}(X)$, the distributive laws are exactly the set identities $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ from Theorem 3.22. Distributivity is the condition that separates lattices that "look like sets" from those that do not (e.g., the lattice of closed subspaces of a Hilbert space is not distributive).

Definition (Complement and Boolean Lattice)

Definition 4.4.4 (Complement and Boolean Lattice)**.** Let $(L, \leq)$ be a bounded distributive lattice with top element $\top$ and bottom element $\bot$. An element $a' \in L$ is a *complement* of $a \in L$ if

$$a \vee a' = \top \quad \text{and} \quad a \wedge a' = \bot.$$

A *Boolean lattice* (or *Boolean algebra*) is a bounded distributive lattice in which every element has a complement.

Remark (The power set is the canonical Boolean lattice)*.* In $\mathcal{P}(X)$, the complement of a set A is $A^c = X \setminus A$, since $A \cup A^c = X = \top$ and $A \cap A^c = \varnothing = \bot$. The power set is therefore a complete Boolean lattice. Every σ-algebra on X is a Boolean sub-algebra of $\mathcal{P}(X)$: it is closed under complements (by definition) and under finite meets and joins (since every σ-algebra is an algebra).

Remark (σ-algebras as sub-lattices)*.* A σ-algebra $\mathcal{M}$ on X is, from the lattice-theoretic viewpoint, a sub-structure of $(\mathcal{P}(X), \subseteq)$ that is:

- a complete sub-lattice (closed under all countable joins and meets);
- a Boolean sub-algebra (closed under complementation).

The condition "closed under countable joins" is exactly the σ-algebra axiom that countable unions of measurable sets are measurable. The condition "closed under meets" for countable families follows from De Morgan together with closure under countable unions and complements.

This perspective makes the definition of σ-algebra feel natural: it is exactly a complete Boolean sub-algebra of the power set lattice, where "complete" is interpreted countably (countable joins and meets), not set-theoretically.

Remark (Topologies as join-sub-structures). A topology τ on X is *not* a sub-lattice of $\mathcal{P}(X)$ in general, because it is not closed under arbitrary meets (i.e., arbitrary intersections of open sets need not be open). However, τ is closed under arbitrary joins (arbitrary unions of open sets are open) and finite meets (finite intersections of open sets are open). This makes τ an *inf-semilattice* for finite meets and a complete *join-semilattice*. The failure of closure under infinite meets is fundamental: it is precisely what allows open sets to "separate" points, and it is the reason topology is structurally different from measure theory.

Theorem (Knaster–Tarski Fixed-Point Theorem)

Let $(L, \leq)$ be a complete lattice and $f : L \to L$ a monotone function (i.e., $a \leq b \Rightarrow f(a) \leq f(b)$). Then f has a least fixed point and a greatest fixed point in L. Specifically:

$$\mathrm{lfp}(f) = \inf\{\, a \in L : f(a) \leq a \,\}, \qquad \mathrm{gfp}(f) = \sup\{\, a \in L : a \leq f(a) \,\}.$$

Remark (Standard quantified statement). $\forall$ complete lattices $(L, \leq)$, $\forall$ monotone $f : L \to L$: $\exists p, q \in L$ such that $f(p) = p$, $f(q) = q$, and $\forall r \in L$ with $f(r) = r$: $p \leq r \leq q$.

Remark (Why this theorem appears here). The Knaster–Tarski theorem is not needed for most of Volume I, but it is placed here because its hypotheses — complete lattice, monotone function — are now in place and the statement is short. Its significance reaches further:

- In set theory, it is used to prove the Schröder–Bernstein theorem without the Axiom of Choice (the monotone map sends $A \mapsto X \setminus g(Y \setminus f(A))$ on $\mathcal{P}(X)$).
- In measure theory, it is used to construct the smallest σ-algebra containing a given class of sets (the generated σ-algebra is a fixed point of the "closure under countable operations" operator).
- In theoretical computer science and domain theory, it is the foundation for denotational semantics.

For now, note that $(\mathcal{P}(X), \subseteq)$ is a complete lattice and any set operation that is monotone (such as $A \mapsto \overline{A}$, closure in a topological space, or $A \mapsto \sigma(A)$, generated σ-algebra) has a fixed point by this theorem.

Remark (Transition). With the notion of lattice in hand, the Hasse diagram becomes a more powerful visualisation tool: the Hasse diagram of a lattice has a unique bottom element ($\inf L$) and a unique top element ($\sup L$), and the join and meet of any two elements can be read off as the unique path structure in the diagram. The next section introduces Hasse diagrams, the duality principle, and the connection between duality and the passage from sup to inf.

Remark (Dependencies).

- Distributive Lattice
- Complete Lattice

Hasse Diagrams and Duality — Quick Reference

Concept	Meaning	Detail
Cover relation	$a \lessdot b$: $a < b$ with nothing strictly between	↓ Def
Hasse diagram	Graph encoding a poset; edges only for cover relations	↓ Def
Dual poset	$(A, \geq)$ obtained by reversing the order on $(A, \leq)$	↓ Def
Duality principle	Every theorem about posets yields a dual theorem for free	↓ Prop

4.5 Hasse Diagrams and the Duality Principle

A partial order can be an enormous set of pairs, impossible to visualise directly. The *Hasse diagram* is a compression: it retains only the pairs that are "immediate" in the order, discarding those implied by transitivity and reflexivity. From the compressed diagram the full order can be recovered by taking transitive closure. Hasse diagrams are the standard visual language for finite posets; knowing how to read and draw them is essential for understanding the order examples in topology and lattice theory.

Definition (Cover Relation)

Definition 4.5.1 (Cover Relation)**.** Let $(A, \leq)$ be a poset and $a, b \in A$. We say b *covers* a, written $a \lessdot b$, if $a < b$ and there is no $c \in A$ with $a < c < b$.

Remark (Standard quantified statement). $a \lessdot b \ \leftrightarrow\ (a < b) \ \wedge\ \neg(\exists c \in A,\ a < c \ \wedge\ c < b)$.

Remark (Intuition). The cover relation strips the partial order down to its "essential edges." If $a \lessdot b$, then b is the *immediate successor* of a: you cannot insert anything between them. In a finite poset, every strict relation $a < b$ factors as a finite chain of cover steps: $a = a_0 \lessdot a_1 \lessdot \cdots \lessdot a_k = b$. In an infinite poset, this factorisation may fail (e.g., in $(\mathbb{R}, \leq)$, no two distinct real numbers are in a cover relation because there is always a rational strictly between them).

Definition (Hasse Diagram)

Remark (Dependencies).

- Partial Order
- Strict Order

Definition 4.5.2 (Hasse Diagram)**.** The *Hasse diagram* of a finite poset $(A, \leq)$ is the directed graph whose vertices are the elements of A, with an upward edge from a to b whenever $a \lessdot b$. Three graphical conventions apply:

(i) Reflexive loops ($a \leq a$) are suppressed.

(ii) Edges implied by transitivity are suppressed: if $a \lessdot b$ and $b \lessdot c$, no direct edge from a to c is drawn.

(iii) Direction is encoded by height: $a \lessdot b$ is drawn with b strictly above a, so arrowheads are unnecessary.

Remark (Dependencies).

- Partial Order
- Cover Relation
- Finite and Infinite Sets

Remark (How to recover the full order). The full order can be read back from the Hasse diagram: $a \leq b$ if and only if either $a = b$ or there is an upward path in the diagram from a to b (a sequence of cover edges going strictly upward). The diagram is therefore a lossless encoding of the partial order, compressed by removing the redundant reflexive and transitive edges.

Example (Divisibility on $\{1, 2, 3, 4, 6, 12\}$). Order the set $D = \{1, 2, 3, 4, 6, 12\}$ by divisibility ($a \leq b$ iff $a \mid b$). The covers are:

$$1 \lessdot 2, \quad 1 \lessdot 3, \quad 2 \lessdot 4, \quad 2 \lessdot 6, \quad 3 \lessdot 6, \quad 4 \lessdot 12, \quad 6 \lessdot 12.$$

The Hasse diagram places 1 at the bottom and 12 at the top. The relation $1 \leq 4$ holds in the poset but the edge $1 \to 4$ is *not* drawn because $1 \lessdot 2 \lessdot 4$ provides an upward path. Drawing the direct edge would be redundant.

Notice also: 4 and 6 are incomparable, so there is no edge between them — they sit at the same level with no path connecting them. The join $4 \vee 6 = 12 = \operatorname{lcm}(4, 6)$ and the meet $4 \wedge 6 = 2 = \gcd(4, 6)$ are readable from the diagram as the lowest common ancestor and the highest common descendant, respectively.

Example (Power set $\mathcal{P}(\{a, b, c\})$ under inclusion). The Hasse diagram of $(\mathcal{P}(\{a, b, c\}), \subseteq)$ has four levels: $\varnothing$ at the bottom; the three singletons $\{a\}, \{b\}, \{c\}$ one level up; the three two-element sets $\{a, b\}, \{a, c\}, \{b, c\}$ above those; and $\{a, b, c\}$ at the top. Each singleton is covered by the two two-element sets containing it. This diagram is the Boolean lattice B_3: it has a bottom ($\varnothing$), a top ($\{a, b, c\}$), and every element has a complement ($A' = \{a, b, c\} \setminus A$).

Remark (Limitation to finite posets). For infinite posets such as $(\mathbb{N}, \leq)$, the cover relation is well-defined ($n \lessdot n+1$) but the full diagram cannot be drawn. In $(\mathbb{R}, \leq)$ or $(\mathbb{Q}, \leq)$, the cover relation is empty: no two elements are in a cover relation. Hasse diagrams are a practical visualisation tool for finite or schematic examples and should not be used as a proof device in infinite settings.

Definition (Dual Poset)

Definition 4.5.3 (Dual Poset)**.** Let $(A, \leq)$ be a poset. The *dual poset* is $(A, \geq)$, where $a \geq b$ is defined to mean $b \leq a$. The dual is also written $(A, \leq^{\mathrm{op}})$.

Remark (Dependencies).

- Partial Order
- Relation

Remark (Intuition). The dual poset is obtained by flipping the Hasse diagram upside down. Every structural feature is preserved but reflected: elements at the top are now at the bottom, minimal elements become maximal, upper bounds become lower bounds, and suprema become infima and vice versa (as established in Proposition 4.3.4).

Proposition

Proposition 4.5.4 (Duality Principle for Posets)**.** *Let Φ be any first-order statement about a poset $(A, \leq)$ expressed using only the relation $\leq$. Let Φ^* be the statement obtained from Φ by replacing every occurrence of $\leq$ with $\geq$ (equivalently, working in the dual poset). Then:*

$$(A, \leq) \models \Phi \iff (A, \geq) \models \Phi.$$

In particular, if Φ is a theorem about all posets, then so is Φ^*.

Remark (Standard quantified statement). $\forall$ posets $(A, \leq)$, $\forall$ first-order sentences Φ in the language of $\leq$: $(A, \leq) \models \Phi \leftrightarrow (A, \geq) \models \Phi^*$, where Φ^* substitutes $\geq$ for every $\leq$.

Remark (Dependencies).

- Partial Order
- Relation

Remark (Theorems come in pairs at no extra cost). The duality principle means every proved theorem about posets automatically yields a second theorem — its dual — without any additional work. You will encounter this repeatedly in analysis. Here are the most important instances:

- *Supremum $\leftrightarrow$ Infimum.* Every theorem about sup yields a theorem about inf by duality. In particular, the completeness axiom for $\mathbb{R}$ (every nonempty bounded-above subset has a supremum) immediately implies that every nonempty bounded-below subset has an infimum — not a separate axiom but a consequence by duality.

- *Minimal ↔ Maximal.* A statement about minimal elements dualises to a statement about maximal elements. Zorn's Lemma (every poset in which every chain is bounded above has a maximal element) dualises to: every poset in which every chain is bounded below has a minimal element.

- *Well-ordering is not self-dual.* $(\mathbb{N}, \leq)$ is well-ordered (every nonempty subset has a least element); its dual $(\mathbb{N}, \geq)$ is not well-ordered because the whole set has no greatest element. Duality preserves the form of a definition but not whether a particular structure satisfies it.

- *Join ↔ Meet in lattices.* Every lattice identity involving $\vee$ and $\wedge$ dualises by swapping the two operations. The distributive law $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ dualises to $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$: proving one gives the other for free.

Remark (Transition). The order vocabulary developed across the order sections — partial orders, total orders, well-orders, bounds, suprema, infima, lattices, directed sets, and duality — is the complete toolkit for order-theoretic reasoning in analysis. In Volume II, the construction of $\mathbb{R}$ requires only the completeness axiom (existence of suprema) and the Dedekind cut or Cauchy sequence constructions, both of which use only the vocabulary established here. In Volume III, metric space arguments, the definition of compactness, and the construction of Lebesgue measure all draw on this vocabulary repeatedly.

Order Extensions — Quick Reference

Concept	**Meaning**	**Detail**
Preorder	Reflexive, transitive (not necessarily antisymmetric)	↓ Def
Loset	Linear order = complete partial order	↓ Def
Symmetric part	$xIy \iff xRy \wedge yRx$	↓ Def
Asymmetric part	$xPy \iff xRy \wedge \neg(yRx)$	↓ Def
Transitive closure	Smallest transitive relation containing R	↓ Def
Extension	$\succsim'$ agrees with $\succsim$ on strict and weak ranks	↓ Def
OWC	$xT(\succsim)y \Rightarrow \neg(y \succ x)$	↓ Def
Szpilrajn's Thm	Every partial order extends to a linear order	↓ Thm

4.6 Order Extensions

Definition (Symmetric and Asymmetric Parts)

Definition 4.6.1 (Symmetric and Asymmetric Parts)**.** For any binary relation R on X, define:

- the *symmetric part*: $xIy \iff xRy$ and yRx
- the *asymmetric part*: $xPy \iff xRy$ but not yRx

Note that $R = P \cup I$.

Remark (Dependencies).

- Relation

- Union
- Set Difference

Remark (Preference-theoretic reading). In decision theory, R encodes a preference relation: xRy means "x is at least as good as y." Then I is the *indifference* relation (equally good) and P is the *strict preference* relation (strictly better). This decomposition is fundamental to utility representation theory.

Definition (Preorder and Loset)

A binary relation R on X is a *preorder* if it is reflexive and transitive.

A *loset* (linearly ordered set) is a pair (X, R) where R is a *linear order*: a complete partial order (R is reflexive, transitive, antisymmetric, and complete).

Remark (Preorder vs. partial order). A preorder need not be antisymmetric: two distinct elements may satisfy xRy and yRx simultaneously (they are "indifferent" but not equal). Adding antisymmetry promotes a preorder to a partial order. Every partial order is a preorder, but not conversely.

Definition (Maximal Elements and Upper Bounds)

Definition 4.6.2 (Maximal Elements and Upper Bounds)**.** Let $\succsim$ be a binary relation on X.

- The set of *maximal elements* of X is
$$\mathrm{Max}(X, \succsim) = \{x \in X \mid y \succsim x \text{ for no } y \in X \text{ with } y \neq x\}.$$
- The set of *upper bounds* of X is
$$M(X, \succsim) = \{x \in X \mid x \succsim y \text{ for all } y \in X\}.$$

Remark (Dependencies).

- Relation
- Upper and Lower Bounds
- Strict Order

Remark (Maximal vs. greatest). A maximal element has nothing strictly above it; an upper bound (greatest element) is above everything. In a partial order, the greatest element is unique if it exists and is always maximal, but maximal elements need not be greatest. In a linear order, maximal and greatest coincide.

Definition (Transitive Closure)

Definition 4.6.3 (Transitive Closure)**.** The *transitive closure* $T(R)$ of a binary relation R on X is the smallest transitive relation containing R: that is, $T(R)$ is transitive, $xRy \Rightarrow xT(R)y$, and no strictly smaller relation is both transitive and contains R.

Remark (Dependencies).

- Relation
- Transitive Relation
- Intersection

Remark (Constructive description). Define $R^0 = R$ and $xR^m y$ if there exist $z_1, \ldots, z_m \in X$ such that $xRz_1R\cdots Rz_mRy$. Then

$$T(R) = R \cup \bigcup_{m\in\mathbb{N}} R^m.$$

Existence follows from the fact that the intersection of any collection of transitive relations is transitive, so the intersection of all transitive supersets of R is the smallest such relation.

Definition (Extension of a Preorder)

Definition 4.6.4 (Extension of a Preorder)**.** Let $\succsim$ be a preorder on X. A preorder $\succsim'$ is an *extension* of $\succsim$ if

$$x \succsim y \;\Rightarrow\; x \succsim' y, \qquad x \succ y \;\Rightarrow\; x \succ' y.$$

A *complete extension* is an extension that is also complete.

Remark (Dependencies).

- Preorder
- Relation

Remark (What extensions do). An extension of $\succsim$ "fills in" the comparisons left undecided by $\succsim$ without reversing any existing strict comparison. The goal is to promote a partial ranking to a total ranking while respecting the original judgements.

Theorem (Szpilrajn, 1930)

For any nonempty set X and partial order $\succsim$ on X, there exists a linear order that is an extension of $\succsim$.

Remark (Proof sketch via Hausdorff Maximum Principle). Let T_X be the set of all partial orders on X extending $\succsim$, ordered by inclusion. By the Hausdorff Maximum Principle (equivalent to AC), T_X has a maximal chain; its union $\succsim^*$ is a partial order extending $\succsim$. If $\succsim^*$ were not complete, one could enlarge it via transitive closure, contradicting maximality. Hence $\succsim^*$ is a linear order extending $\succsim$.

Dependence on AC. The theorem is equivalent to the Axiom of Choice for infinite sets; no proof avoiding AC is known.

Corollary (Complete preorder extension)

Corollary 4.6.5 (Complete preorder extension)**.** *For any nonempty set X and preorder $\succsim$ on X, there exists a complete preorder that is an extension of $\succsim$.*

Remark (Proof). Proof details.

Remark (Standard quantified statement). $\forall X \neq \varnothing$, $\forall \succsim$ preorder on X: $\exists \succsim'$ complete preorder on X s.t. $(x \succsim y \to x \succsim' y) \wedge (x \succ y \to x \succ' y)$.

Remark (Dependencies).

- Preorder
- Extension of a Preorder
- Quotient Set
- Symmetric and Asymmetric Parts

Remark (Proof). Pass to the quotient $X/\sim$ (where $\sim$ is the symmetric part), apply Szpilrajn's Theorem to obtain a linear order on $X/\sim$, then pull back to a complete preorder on X.

Definition and Proposition (Only Weak Cycles)

Definition 4.6.6 (and Proposition (Only Weak Cycles))**.** Let $\succsim$ be a binary relation on X with asymmetric part $\succ$ and transitive closure $T(\succsim)$. We say $\succsim$ satisfies *only weak cycles* (OWC) if

$$xT(\succsim)y \;\Rightarrow\; \neg(y \succ x).$$

Proposition. A binary relation $\succsim$ on a nonempty set X can be extended to a complete preorder if and only if it satisfies OWC.

Remark (Dependencies).

- Relation
- Symmetric and Asymmetric Parts
- Transitive Closure
- Extension of a Preorder

Remark (Intuition for OWC). OWC prohibits the following: there is a "chain" of weak preferences $x_1 \succsim x_2 \succsim \cdots \succsim x_n$ but a strict reversal $x_n \succ x_1$. Such a configuration cannot be extended to a complete preorder because the chain forces $x_1 \succsim' x_n$ in any extension, but strict preference $x_n \succ x_1$ in the extension would mean $x_n \succsim' x_1$ but not $x_1 \succsim' x_n$ — contradicting $x_1 \succsim' x_n$.

Remark (Transition to AC). Szpilrajn's Theorem relies on Zorn's Lemma / Hausdorff Maximum Principle, which are equivalent to the Axiom of Choice. The next section develops these equivalences directly.

Induced Orders and Order Embeddings — Quick Reference

Concept	Meaning	Detail
Induced preorder	$x \leq_f y \Leftrightarrow f(x) \leq' f(y)$; always a preorder	↓ Def
Induced partial order	Induced order is a partial order iff f is injective	↓ Prop
f-indistinguishable	$x \sim_f y$: both $x \leq_f y$ and $y \leq_f x$	↓ Def
Order embedding	Injective, order-preserving, and order-reflecting	↓ Def
Embedding vs. isomorphism	Embedding is iso onto image; isomorphism is surjective too	↓ Prop
Suborder	Restriction of $\leq'$ to a subset $S \subseteq B$	↓ Def
Key results:		
$\leq_f$ preorder: always. Antisymmetry: iff f injective.		↓
f order embedding $\Rightarrow (A, \leq_f) \cong (f(A), \leq')$.		↓

4.7 Induced Orders and Order Embeddings

The simplest way to put an order on a set A is to compare elements of A *indirectly* through a function into an already-ordered set. This construction, called the *induced order* or *pullback order*, is ubiquitous: it explains how the usual order on $\mathbb{Z}$ restricts to $\mathbb{N}$, how functions on a common domain acquire a pointwise order, and how subsets inherit the order of the ambient space.

Definition (Induced Order)

Definition 4.7.1 (Induced Order)**.** Let $(B, \leq')$ be a partially ordered set and let $f : A \to B$ be a function. The *order induced by* f (or *pullback order along* f) is the relation $\leq_f$ on A defined by:

$$x \ \leq_f \ y \quad \Longleftrightarrow \quad f(x) \ \leq' \ f(y).$$

Remark (Intuition). Two elements of A are compared by sending them through f and comparing their images in $(B, \leq')$. The order on A is entirely inherited from B via f: we never compare elements of A directly, only their f-values.

Remark (Standard quantified statement). $\forall x, y \in A, \quad x \leq_f y \iff f(x) \leq' f(y)$.

Remark (Dependencies).

- Partial Order
- Function
- Relation

Proposition

Proposition 4.7.2 ($\leq_f$ is always a preorder)**.** *For any function $f : A \to B$ and partial order $(B, \leq')$, the relation $\leq_f$ is a preorder on A: it is reflexive and transitive.*

Remark (Standard quantified statement). $\forall x \in A : \ f(x) \leq' f(x)$ (reflexivity); $\forall x, y, z \in A : \ f(x) \leq' f(y) \wedge f(y) \leq' f(z) \to f(x) \leq' f(z)$ (transitivity).

Remark (Dependencies).

- Induced Order
- Preorder
- Function
- Partial Order

Remark (Proof strategy). Both properties are inherited directly from $(B, \leq')$ by pulling back through f: reflexivity at x uses reflexivity at $f(x)$, and transitivity along x, y, z uses transitivity along $f(x), f(y), f(z)$.

Definition (f-Indistinguishable Elements)

Definition 4.7.3 (f-Indistinguishable Elements)**.** Let $f : A \to B$ and $\leq_f$ be the induced order. Two elements $x, y \in A$ are *f-indistinguishable*, written $x \sim_f y$, if both $x \leq_f y$ and $y \leq_f x$, i.e. if

$$f(x) \leq' f(y) \quad \text{and} \quad f(y) \leq' f(x).$$

Remark (Dependencies).

- Induced Order
- Function
- Partial Order

Remark (Intuition). Two elements are f-indistinguishable when they map to the same position in the $\leq'$-order — not necessarily to the same *point*, but to mutually comparable points. When $\leq'$ is a partial order (antisymmetric), $x \sim_f y$ forces $f(x) = f(y)$, making f non-injective the only obstruction to antisymmetry.

Proposition

Proposition 4.7.4 ($\leq_f$ is a partial order iff f is injective)**.** *Let $(B, \leq')$ be a partially ordered set and $f : A \to B$. The induced order $\leq_f$ is a partial order on A if and only if f is injective.*

Remark (Standard quantified statement). $\leq_f$ is a partial order $\leftrightarrow$ $\forall x, y \in A,\ f(x) \leq' f(y) \wedge f(y) \leq' f(x) \to x = y$ $\leftrightarrow$ $\forall x, y \in A,\ f(x) = f(y) \to x = y$ (injectivity of f, since $\leq'$ is antisymmetric).

Remark (Dependencies).

- Induced Order
- Partial Order
- Injective Function

Remark (Common error). It is tempting to think the induced order is automatically a partial order "because $\leq'$ is one." It is not. The issue is antisymmetry: if f collapses two distinct points $x \neq y$ to the same or comparable images, we get $x \leq_f y$ and $y \leq_f x$ with $x \neq y$, violating antisymmetry. Reflexivity and transitivity lift freely; antisymmetry does not.

Example (Non-injective f destroys antisymmetry). Let $B = (\mathbb{N}, \leq)$ and let $f : \{a, b\} \to \mathbb{N}$ be the constant function $f(a) = f(b) = 0$. Then $a \leq_f b$ (since $0 \leq 0$) and $b \leq_f a$ (since $0 \leq 0$), but $a \neq b$. So $\leq_f$ is not antisymmetric, confirming that non-injectivity forces a failure.

Example (Injective f gives a genuine partial order). Let $B = (\mathcal{P}(\{1, 2, 3\}), \subseteq)$ and let $A = \{x, y, z\}$ with $f(x) = \{1\}$, $f(y) = \{2\}$, $f(z) = \{1, 2\}$. Then f is injective. The induced order has $x \leq_f z$ (since $\{1\} \subseteq \{1, 2\}$) and $y \leq_f z$ (since $\{2\} \subseteq \{1, 2\}$), but x and y are incomparable. This is a genuine partial order on A.

Definition (Suborder / Restriction)

Definition 4.7.5 (Suborder / Restriction)**.** Let $(B, \leq')$ be a partially ordered set and $S \subseteq B$. The *suborder* on S (or *induced order on* S) is the relation $\leq'_S$ defined by:

$$x \ \leq'_S \ y \quad \Longleftrightarrow \quad x \leq' y, \qquad \text{for } x, y \in S.$$

Remark (Dependencies).

- Partial Order
- Subset
- Relation

Remark (Connection to induced orders). The suborder on $S \subseteq B$ is exactly the order induced by the inclusion map $\iota : S \hookrightarrow B$, $\iota(x) = x$. Since ι is injective, the suborder is always a partial order whenever $(B, \leq')$ is. This justifies speaking of "S with the inherited order" without further verification.

Definition (Order Embedding)

Definition 4.7.6 (Order Embedding)**.** Let $(A, \leq)$ and $(B, \leq')$ be partially ordered sets. A function $f : A \to B$ is an *order embedding* if for all $x, y \in A$:

$$x \ \leq \ y \quad \Longleftrightarrow \quad f(x) \ \leq' \ f(y).$$

Remark (Dependencies).

- Ordered Set
- Function

Remark (Two directions, one condition). The $(\Rightarrow)$ direction says f is order-preserving (see Definition (Order-Preserving Map)). The $(\Leftarrow)$ direction — $f(x) \leq' f(y) \Rightarrow x \leq y$ — is called *order-reflection*. An order embedding both preserves and reflects the order, so f is a perfect local copy of $(A, \leq)$ inside $(B, \leq')$.

Remark (Relation to the induced order). If $f : A \to B$ is an order embedding from $(A, \leq)$ to $(B, \leq')$, then $\leq$ coincides with the induced order $\leq_f$:

$$x \leq y \iff f(x) \leq' f(y) \iff x \leq_f y.$$

In other words, an order embedding is exactly an injective function whose induced order *agrees with the existing order on* A.

Proposition (Order embeddings are injective)

Proposition 4.7.7 (Order embeddings are injective)**.** *Every order embedding* $f : (A, \leq) \to (B, \leq')$ *is injective. Go to proof.*

Remark (Standard quantified statement). $\forall x, y \in A : \; f(x) = f(y) \to x = y$.

Remark (Dependencies).

- Order Embedding
- Injective Function
- Partial Order

Remark (Proof strategy). Injectivity falls out of the reflection direction: if two elements have the same image, reflection forces them to be mutually $\leq$-related, and antisymmetry collapses them to equality.

Proposition (Order embedding is isomorphism onto image)

Proposition 4.7.8 (Order embedding is isomorphism onto image)**.** *If* $f : (A, \leq) \to (B, \leq')$ *is an order embedding, then* f *is an order isomorphism from* $(A, \leq)$ *to the suborder* $(f(A), \leq'_{f(A)})$*. Go to proof.*

Remark (Standard quantified statement). $\forall x, y \in A : \; x \leq y \leftrightarrow f(x) \leq'_{f(A)} f(y)$, i.e. f is an order isomorphism $(A, \leq) \cong (f(A), \leq')$.

Remark (Dependencies).

- Order Embedding
- Image of a Set
- Suborder
- Partial Order

Remark (Consequence). This proposition is the order-theoretic analogue of the first isomorphism theorem: an order embedding always realises $(A, \leq)$ as an isomorphic copy of a sub-poset of $(B, \leq')$. When f is also surjective, the embedding becomes a full order isomorphism (see Definition (Order Isomorphism)).

Example (Standard embeddings). Each of the following is an order embedding:

(i) The inclusion $\mathbb{N} \hookrightarrow \mathbb{Z}$ with standard $\leq$ on both: $m \leq n \iff m \leq n$ trivially. The natural numbers sit inside the integers as an isomorphic copy of $(\mathbb{N}, \leq)$.

(ii) The function $f : \mathbb{N} \to \mathbb{N}$, $f(n) = 2n$, embeds $(\mathbb{N}, \leq)$ into itself: $m \leq n \iff 2m \leq 2n$. The image is the even numbers with the inherited order.

(iii) The map $A \mapsto A$ from $(\mathcal{P}(X), \subseteq)$ to itself is an order isomorphism (the identity), but any injective $g : \mathcal{P}(X) \to \mathcal{P}(X)$ that preserves and reflects $\subseteq$ is an order embedding.

Remark (Transition). Induced orders and embeddings are the order-theoretic counterpart of substructures in algebra. Just as a subgroup is a subset closed under the group operations, a sub-poset is a subset with the inherited order — and an order embedding is the morphism that witnesses this. In analysis, these ideas appear when restricting an order from $\mathbb{R}$ to subsets (intervals, sequences, function spaces), and when the completeness property is transferred between ordered structures.

4.8 Zorn

AC Equivalences — Quick Reference

Statement	Role
Axiom of Choice (AC)	Choice function from any collection of nonempty sets
Product form of AC	Cartesian product of nonempty sets is nonempty
Zorn's Lemma	Poset with every chain bounded $\Rightarrow$ maximal element
Hausdorff Maximum Principle	Every poset has a $\subseteq$-maximal chain
Well-Ordering Theorem	Every set admits a well-ordering

All five statements are equivalent (over ZF set theory).

4.9 Zorn's Lemma and the Axiom of Choice

Remark (Why AC is non-constructive). For finite collections, the axiom is provable without extra assumptions (see SRF-STO-C08-S8.1). For infinite collections, no construction can produce the choice function: AC asserts its existence without exhibiting it. This non-constructive character is both AC's power and the source of the foundational controversy surrounding it.

Zorn's Lemma

Let $(P, \leq)$ be a poset in which every chain (loset) has an upper bound in P. Then P has at least one maximal element.

Remark (Reading Zorn's Lemma). The hypothesis "every chain has an upper bound" does *not* require the bound to lie in the chain itself. The conclusion gives a maximal element: an element above which nothing in P sits. In a partial order, there may be many maximal elements; Zorn guarantees at least one.

Remark (Typical proof pattern using Zorn's Lemma). To show an object of type X exists with a maximal property:

1. Partially order candidate objects by inclusion (or extension).
2. Verify every chain of candidates has an upper bound (usually its union).

3. Conclude by Zorn that a maximal candidate exists.
4. Show the maximal candidate has the desired property (often: if it lacked it, it could be enlarged, contradicting maximality).

This pattern recurs throughout algebra and analysis: maximal ideals, algebraic bases (Hamel bases), maximal consistent sets, and Szpilrajn's Theorem.

Hausdorff Maximum Principle

In every poset $(P, \leq)$ there exists a $\subseteq$-maximal chain; that is, a chain $C \subseteq P$ such that no chain $C' \subseteq P$ strictly contains C.

Remark (Relation to Zorn). The Hausdorff Maximum Principle is an immediate corollary of Zorn's Lemma: take the poset of chains ordered by inclusion; every chain of chains is bounded by its union; Zorn gives a maximal chain. Conversely, Hausdorff implies Zorn. Both are equivalent to AC over ZF.

Remark (Application: Szpilrajn via Hausdorff). In the proof of Szpilrajn's Theorem (↑ Thm), let T_X be the set of all partial orders on X extending $\succsim$, ordered by $\subseteq$. By the Hausdorff Maximum Principle, T_X has a maximal chain A; its union $\succsim^* = \bigcup A$ is a partial order extending $\succsim$. If $\succsim^*$ were not total, there would exist incomparable x, y and one could extend $\succsim^*$ by adding (x, y) (closing under transitivity), contradicting maximality of A. Hence $\succsim^*$ is a linear order. $\square$

Remark (Transition). Zorn's Lemma is the workhorse for existence proofs throughout abstract algebra and analysis. It first appears in the project context here and at the proof stubs in §11 of the Chapter VIII exercises.

Chapter 5

Cardinality

cardinality

Orderings $\rightarrow$ **Cardinality** $\rightarrow$ Proof Techniques

5.1 Cardinality

Cardinality — Quick Reference

Concept	Meaning	Detail
Same cardinality	Bijection $f : A \to B$ exists; written $A \sim B$	↓ Def
Finite set	$A \sim \{1, \ldots, n\}$ for some $n \in \mathbb{N}$, or $A = \varnothing$	↓ Def
Countably infinite	$A \sim \mathbb{N}$	↓ Def
Countable	Finite or countably infinite	↓ Def
Uncountable	Infinite but not countable	↓ Def
$\lvert A\rvert \leq \lvert B\rvert$	Injection $A \hookrightarrow B$ exists	↓ Def
$\mathbb{Q}$ countable	Diagonal enumeration	↓ Thm
$\mathbb{R}$ uncountable	Cantor diagonal argument	↓ Thm
Schröder–Bernstein	$\lvert A\rvert \leq \lvert B\rvert$ and $\lvert B\rvert \leq \lvert A\rvert \Rightarrow A \sim B$	↓ Thm
Cantor's theorem	$\lvert A\rvert < \lvert\mathcal{P}(A)\rvert$ for every set A	↓ Thm
Countable union	Countable union of countable sets is countable	↓ Thm

5.2 Cardinality and Infinite Sets

The intuitive idea of "size" works perfectly for finite sets: $\{a, b, c\}$ has three elements, $\{1, 2, 3, 4\}$ has four, and any two finite sets can be compared by counting. For infinite sets, counting breaks down, and a different foundation is needed. The key insight, due to Cantor, is that two sets have the "same size" if and only if their elements can be matched perfectly — one-to-one and onto — without any element left over on either side. This single idea, made precise through the notion of a bijection, gives a theory of size that applies uniformly to both finite and infinite sets, and produces genuinely surprising results: the integers and the rationals turn out to have the same size as the natural numbers, while the real numbers are provably larger.

Definition (Same Cardinality)

Definition 5.2.1 (Same Cardinality)**.** Two sets A and B have the *same cardinality*, written $A \sim B$, if there exists a bijection $f : A \to B$.

Remark (Standard quantified statement). $A \sim B \leftrightarrow \exists f : A \to B, \ (\forall a_1, a_2 \in A, \ f(a_1) = f(a_2) \to a_1 = a_2) \wedge (\forall b \in B, \ \exists a \in A, \ f(a) = b)$.

Remark (Dependencies).

- Bijective Function

Remark (Same cardinality is an equivalence relation). The relation $\sim$ is an equivalence relation on the class of all sets:

(i) *Reflexive*: $A \sim A$ via id_A.

(ii) *Symmetric*: if $f : A \to B$ is a bijection then $f^{-1} : B \to A$ is a bijection, so $A \sim B$ implies $B \sim A$.

(iii) *Transitive*: if $f : A \to B$ and $g : B \to C$ are bijections then $g \circ f : A \to C$ is a bijection, so $A \sim B$ and $B \sim C$ imply $A \sim C$.

This is a genuine equivalence relation, so all the theory of equivalence classes developed in the previous section applies. The equivalence class of A under $\sim$ is called the *cardinality* of A, written $|A|$.

Example (Infinite sets can be equinumerous with proper subsets). Let $E = \{2, 4, 6, \dots\}$ be the set of positive even integers. The function $f : \mathbb{N} \to E$ defined by $f(n) = 2n$ is a bijection, so $\mathbb{N} \sim E$. This seems paradoxical: E is a proper subset of $\mathbb{N}$, yet the two sets have the same cardinality. This is not a contradiction but rather one of the defining characteristics of infinite sets — a phenomenon that simply cannot occur for finite sets.

Remark (Dedekind's characterization of infinite sets). The phenomenon in Example 5.2 is not an accident. An infinite set can always be matched bijectively with some proper subset of itself. This is sometimes taken as the *definition* of an infinite set (Dedekind's definition): a set is infinite if and only if it is equinumerous with a proper subset of itself.

Definition (Finite and Infinite Sets)

Definition 5.2.2 (Finite and Infinite Sets)**.** A set A is *finite* if $A = \varnothing$ or $A \sim \{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$. A set is *infinite* if it is not finite.

Remark (Standard quantified statement). A is finite $\leftrightarrow$ $A = \varnothing \vee \exists n \in \mathbb{N}, \ \exists f : A \to \{1, \dots, n\}, \ f$ bijective. A is infinite $\leftrightarrow$ $\neg(A$ is finite$)$.

Remark (Dependencies).

- Empty Set
- Same Cardinality
- Natural Numbers

Definition (Countable and Uncountable Sets)

Definition 5.2.3 (Countable and Uncountable Sets)**.** A set A is *countably infinite* if $A \sim \mathbb{N}$. A set is *countable* if it is finite or countably infinite. A set is *uncountable* if it is infinite but not countable.

Remark (Standard quantified statement)*.* A is countably infinite $\leftrightarrow$ $\exists f : A \to \mathbb{N}$, f bijective. A is countable $\leftrightarrow$ A finite $\vee$ $A \sim \mathbb{N}$. A is uncountable $\leftrightarrow$ A infinite $\wedge \neg(A \sim \mathbb{N})$.

Remark (Dependencies)*.*

- Finite and Infinite Sets
- Same Cardinality
- Natural Numbers

Remark (Countable means listable)*.* A set A is countably infinite if and only if its elements can be arranged into an infinite list $a_1, a_2, a_3, \ldots$ with no repetitions and every element of A appearing somewhere in the list. The bijection $f : \mathbb{N} \to A$ is precisely such a listing: $a_n = f(n)$. This is why countability is often informally described as "can be listed."

Definition (Comparing Cardinalities)

Definition 5.2.4 (Comparing Cardinalities)**.** Write $|A| \leq |B|$ if there exists an injection $f : A \hookrightarrow B$. Write $|A| < |B|$ if $|A| \leq |B|$ but $A \not\sim B$.

Remark (Standard quantified statement)*.* $|A| \leq |B|$ $\leftrightarrow$ $\exists f : A \to B$, $\forall a_1, a_2 \in A$, $f(a_1) = f(a_2) \to a_1 = a_2$ (injection exists).

Remark (Dependencies)*.*

- Injective Function
- Same Cardinality

Remark (Why injections, not surjections)*.* An injection $A \hookrightarrow B$ places a "copy" of A inside B without overlap, which formalizes "A is no larger than B." The direction matters: an injection from A to B is not the same as an injection from B to A.

Theorem ($\mathbb{Q}$ is countable)

Theorem 5.2.5 ($\mathbb{Q}$ is countable)**.** *The set $\mathbb{Q}$ of rational numbers is countable.* *Go to proof.*

Remark (Dependencies)*.*

- Countable Set
- Natural Numbers
- Bijective Function

Remark (Proof strategy — diagonal enumeration). The argument arranges all positive rationals into an infinite two-dimensional array: place p/q (in lowest terms, $p, q > 0$) in row p, column q. Then traverse the array along successive diagonals — the anti-diagonal $p + q = k$ for $k = 2, 3, 4, \ldots$ — skipping any fraction already seen. This produces an explicit list of all positive rationals, and one then interleaves the negative rationals and zero. The precise details are left as a guided proof exercise. The core idea is: a doubly infinite list can be converted into a singly infinite list by reading diagonals.

Remark (What this means). $\mathbb{Q}$ is dense in $\mathbb{R}$ — between any two real numbers there is a rational — yet $\mathbb{Q}$ is countable. This means the rationals are in some precise sense "thin" even though they are everywhere. The uncountability of $\mathbb{R}$ that follows shows the reals are genuinely larger.

Theorem (Countable union of countable sets is countable)

Theorem 5.2.6 (Countable union of countable sets is countable)**.** *Let* $\{A_n\}_{n\in\mathbb{N}}$ *be a countable family of countable sets. Then* $\bigcup_{n=1}^{\infty} A_n$ *is countable.* *Go to proof.*

Remark (Dependencies).

- Countable Set
- Natural Numbers
- Indexed Family of Sets
- Union

Remark (Proof strategy). The argument again uses a diagonal enumeration. Arrange the elements of each A_n as a row in an infinite array (padding with repetitions if A_n is finite), then read the array along anti-diagonals. This produces a surjection from $\mathbb{N}$ onto $\bigcup A_n$, which is enough to establish countability. The formal proof is a standard exercise.

Remark (Why this theorem matters downstream). This result is used constantly. In topology: a countable union of nowhere-dense sets is called a *meager* set, and Baire's theorem says a complete metric space is not meager — the proof requires knowing that countable unions behave predictably. In measure theory: a countable union of sets of measure zero has measure zero, by countable additivity. Whenever an argument says "take the union over all $n \in \mathbb{N}$" and needs the result to remain manageable in size, this theorem is the justification.

Theorem ($\mathbb{R}$ is uncountable)

Theorem 5.2.7 ($\mathbb{R}$ is uncountable)**.** *The set* $\mathbb{R}$ *of real numbers is not countable.* *Go to proof.*

Remark (Dependencies).

- Countable Set
- Finite and Infinite Sets
- Countable Union of Countable Sets

Remark (Proof strategy — Cantor's diagonal argument)**.** This is one of the most important proof techniques in all of mathematics. Suppose, for contradiction, that $\mathbb{R}$ were countable, so its elements could be listed: $r_1, r_2, r_3, \ldots$. Write each r_n in decimal expansion. Construct a new real number x by choosing the n-th decimal digit of x to differ from the n-th decimal digit of r_n (and avoiding 9 to prevent the $0.999\cdots = 1$ ambiguity). Then $x \neq r_n$ for every n, because x and r_n differ in the n-th decimal place. So x is not on the list — contradicting the assumption that the list contains every real number.

The key move is the diagonal construction: to defeat every row r_n simultaneously, make x disagree with row n at position n. This "diagonal sweep" is a technique that reappears in logic (Gödel's incompleteness theorem), computability theory (the halting problem), and the proof of Cantor's theorem below.

Remark (Consequence)**.** Since $\mathbb{Q}$ is countable and $\mathbb{R}$ is uncountable, the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ must be uncountable: if they were countable, then $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ would be a countable union of countable sets, hence countable — contradiction.

Theorem (Cantor's Theorem)

Theorem 5.2.8 (Cantor's Theorem)**.** *For any set A, there is no surjection $A \to \mathcal{P}(A)$. In particular, $|A| < |\mathcal{P}(A)|$.* *Go to proof.*

Remark (Standard quantified statement)**.** $\forall A : \neg\exists f : A \to \mathcal{P}(A),\ \forall S \in \mathcal{P}(A),\ \exists a \in A,\ f(a) = S$. Equivalently: $\forall f : A \to \mathcal{P}(A),\ \exists D \in \mathcal{P}(A),\ \forall a \in A,\ f(a) \neq D$.

Remark (Dependencies)**.**

- Function
- Surjective Function
- Power Set
- Axiom Schema of Separation

Remark (Proof strategy — the same diagonal idea)**.** Suppose $f : A \to \mathcal{P}(A)$ is any function. Construct the set

$$D = \{\, a \in A \mid a \notin f(a) \,\}.$$

D is a well-defined subset of A, so $D \in \mathcal{P}(A)$. Claim: D is not in the image of f. For suppose $D = f(a_0)$ for some $a_0 \in A$. Then:

- if $a_0 \in D$, then by definition $a_0 \notin f(a_0) = D$ — contradiction.
- if $a_0 \notin D$, then by definition $a_0 \in f(a_0) = D$ — contradiction.

Either case is impossible, so $D \neq f(a_0)$ for every $a_0 \in A$, and f is not surjective. Since f was arbitrary, no surjection $A \to \mathcal{P}(A)$ exists.

Notice the structure: D is defined by "$a \in D$ iff $a \notin f(a)$," which is a diagonal construction — the membership of a in D is decided by looking at position (a, a) in the "membership table" of f, and then disagreeing. This is the abstract form of the same move used in the proof that $\mathbb{R}$ is uncountable.

Remark (A hierarchy of infinities). Cantor's theorem implies there is no largest cardinality: for any set A, the power set $\mathcal{P}(A)$ is strictly larger. Starting from $\mathbb{N}$, the sets

$$\mathbb{N}, \quad \mathcal{P}(\mathbb{N}), \quad \mathcal{P}(\mathcal{P}(\mathbb{N})), \quad \ldots$$

form a strictly increasing sequence of cardinalities. One can show $|\mathcal{P}(\mathbb{N})| = |\mathbb{R}|$ via the characteristic function bijection (described below), so the second level is already the cardinality of the continuum.

Definition (Function Space B^A)

Definition 5.2.9 (Function Space B^A)**.** For sets A and B, the *function space* B^A denotes the set of all functions from A to B:

$$B^A := \{ f : A \to B \}.$$

Remark (Dependencies).

- Function
- Set and Membership

Remark (Why the exponential notation). For finite sets with $|A| = m$ and $|B| = n$, the number of functions $A \to B$ is n^m (there are n choices for each of the m inputs). The notation B^A reflects this count. In particular, $|\{0,1\}^A| = 2^{|A|}$ for finite A, which matches $|\mathcal{P}(A)| = 2^{|A|}$.

Remark (The characteristic function bijection). For any set A, the power set $\mathcal{P}(A)$ and the function space $\{0,1\}^A$ have the same cardinality via the *characteristic function* bijection:

$$\mathcal{P}(A) \to \{0,1\}^A, \qquad S \mapsto \mathbf{1}_S,$$

where $\mathbf{1}_S(a) = 1$ if $a \in S$ and $\mathbf{1}_S(a) = 0$ if $a \notin S$. This bijection is why Cantor's theorem is sometimes stated as: there is no surjection $A \to \{0,1\}^A$. The notation 2^A is used interchangeably with $\{0,1\}^A$, and $|\mathcal{P}(A)| = 2^{|A|}$ is the general statement.

In functional analysis, spaces of the form B^A — functions from A to B — are the central objects of study. Knowing they are sets (with a definite cardinality when A and B are concrete) is the set-theoretic foundation for everything that follows.

Theorem (Schröder–Bernstein Theorem)

Theorem 5.2.10 (Schröder–Bernstein Theorem)**.** *If $|A| \leq |B|$ and $|B| \leq |A|$, then $A \sim B$.*
Equivalently: if there exist injections $f : A \hookrightarrow B$ and $g : B \hookrightarrow A$, then there exists a bijection $h : A \to B$. *Go to proof.*

Remark (Standard quantified statement). $\big(\exists f : A \hookrightarrow B\big) \wedge \big(\exists g : B \hookrightarrow A\big) \;\to\; \exists h : A \to B,\ h$ bijective.

Remark (Dependencies).

- Comparing Cardinalities

- Injective Function
- Bijective Function

Remark (Why this theorem is indispensable). In practice, proving $A \sim B$ by exhibiting an explicit bijection can be very difficult. Schröder–Bernstein gives a two-injection strategy: find any injection each way, and the theorem guarantees a bijection exists (even if you cannot describe it explicitly). This is a powerful tool for cardinality computations. For example, to show $|(0,1)| = |\mathbb{R}|$, find an injection $(0,1) \hookrightarrow \mathbb{R}$ (inclusion) and an injection $\mathbb{R} \hookrightarrow (0,1)$ (e.g., $x \mapsto \frac{1}{\pi}\arctan(x) + \frac{1}{2}$); the theorem does the rest.

Remark (Proof idea). The proof constructs h by partitioning A into two pieces: those elements whose "ancestry" under repeated application of $g \circ f$ is finite (they go through f), and those whose ancestry is infinite (they go through g^{-1}). The formal argument is a standard exercise; the key insight is that the two injections together cover A in a structured enough way to stitch together a bijection.

Remark (Sequences as functions). A *sequence* in a set X is a function $f : \mathbb{N} \to X$. The element $f(n)$ is written x_n, and the sequence is written $(x_n)_{n\in\mathbb{N}}$ or simply (x_n). This is not a new object: it is an element of the function space $X^{\mathbb{N}}$. When metric spaces and topology discuss convergence of sequences $x_1, x_2, x_3, \ldots$ in an arbitrary space X, the formal object is always a function $\mathbb{N} \to X$, and all the function machinery — injectivity, composition, preimages — applies to it directly.

Remark (Transition). Cardinality distinguishes between the countable (manageable, listable) and the uncountable (genuinely larger). This distinction organizes much of what follows. In metric spaces: separable spaces have countable dense subsets, and separability is a strong structural property. In measure theory: σ-algebras are closed under *countable* unions — not arbitrary ones — precisely because countable operations preserve the kind of control that makes measure theory work. In functional analysis: L^2 spaces have countable orthonormal bases (Hilbert spaces are separable), while non-separable spaces behave very differently. The line between countable and uncountable is not a technicality; it is one of the organizing principles of analysis.

Bibliography

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